

August 25, 2026 04:00 AM GMT

Global Technology

Analog Recap: Margins Overshadow Data Center Commentary

Although cycle recovery is more evident and data center updates offered bright spots the market needed to see, margins were in focus across our global analog coverage. We remain constructive and prefer IFX and STM in Europe, NXP, ADI and ALGM in the US and Renesas, Yangjie and SICC in Asia.

Key Takeaways

- Cycle broadening: Industrial recovery is gaining breadth, while automotive demand is beginning to inflect across the group.
- Margin expectations reset: Q4 margin guides disappointed vs a demanding buy-side bar, particularly at STM and Infineon; the issue is pace of recovery, not demand direction.
- Channel correction largely complete: Distributor inventories are normalising or below target, reducing the risk that current growth is driven by restocking alone.
- Visibility improving: Book-to-bill ratios above 1, firmer pricing and backlog extending through 2026 support a demand-led up-cycle into 2027.
- AI data center remains the structural growth engine: Power semiconductor exposure is scaling rapidly, with raised outlooks and multi-year customer commitments across STM, Texas Instruments, ON Semiconductor and Infineon.

The Q2 2026 reporting season confirms that the analog semiconductor up-cycle is broadening. Across all names, key cycle indicators – channel inventory, pricing, and backlog – are inflecting positively. Drivers for both the autos and industrials end markets are improving and we see strength reported into H2 as near uniform, but with variance in the pace, mix, and impact of each recovery. STM called out c.40% industrials growth yy (Q4), while TI, ON and NXP all pointed to better autos momentum. Generally, however, analog stocks still faced a steep sell-off post earnings as the *bogey trade* prevailed, mainly on margin disappointments. **While we acknowledge weaker margins (Q4 guides primarily), relative to heightened expectations, we think stock reactions this quarter fail to tell the story of improving end-market dynamics and ongoing momentum in the datacenter.**

Margins: Steady but not enough? ... as mentioned, the market had its eye on margins this quarter. While the direction of travel remained positive, the pace of improvement was not always enough to clear a bar that had moved higher through the quarter. **STMicro** was the clearest disappointment. Despite an improving Q2/Q3 gross margin profile (Q3e: c.37% up >200bps qq), the Q4 guide underwhelmed. A

MORGAN STANLEY & CO. INTERNATIONAL PLC+		
Lee Simpson	Equity Analyst	
Lee.Simpson@morganstanley.com		+44 20 7425-3378
MORGAN STANLEY & CO. LLC		
Joseph Moore	Equity Analyst	
Joseph.Moore@morganstanley.com		+1 212 761-7516
MORGAN STANLEY TAIWAN LIMITED+		
Charlie Chan	Equity Analyst	
Charlie.Chan@morganstanley.com		+886 2 2730-1725
MORGAN STANLEY MUFG SECURITIES CO., LTD.+		
Kazuo Yoshikawa, CFA	Equity Analyst	
Kazuo.Yoshikawa@morganstanleymufg.com		+81 3 6836-8408
MORGAN STANLEY & CO. INTERNATIONAL PLC+		
Shawn Kim	Equity Analyst	
Shawn.Kim@morganstanley.com		+44 20 7677-1018
Nigel van Putten	Equity Analyst	
Nigel.Putten@morganstanley.com		+44 20 7425-2803
MORGAN STANLEY ASIA LIMITED+		
Daisy Dai, CFA	Equity Analyst	
Daisy.Dai@morganstanley.com		+852 2848-7310
MORGAN STANLEY & CO. INTERNATIONAL PLC+		
Amelia M Scicluna	Research Associate	
Amelia.Scicluna@morganstanley.com		+44 20 7425-6694
MORGAN STANLEY & CO. LLC		
Nicole Kozhukhov	Research Associate	
Nicole.Kozhukhov@morganstanley.com		+1 212 761-1636
MORGAN STANLEY ASIA LIMITED+		
Henry Zhao	Research Associate	
Henry.Zhao@morganstanley.com		+852 2239-7731
MORGAN STANLEY TAIWAN LIMITED+		
Lucas Wang	Research Associate	
Lucas.Wang@morganstanley.com		+886 2 2730-2875

Morgan Stanley does and seeks to do business with companies covered in Morgan Stanley Research. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of Morgan Stanley Research. Investors should consider Morgan Stanley Research as only a single factor in making their investment decision.

For analyst certification and other important disclosures, refer to the Disclosure Section, located at the end of this report.

+ = Analysts employed by non-U.S. affiliates are not registered with FINRA, may not be associated persons of the member and may not be subject to FINRA restrictions on communications with a subject company, public appearances and trading securities held by a research analyst account.

mix of proforma charges and ramp-up costs mean the hoped-for 40%+ margin looks unachievable, despite a sales guide reaching the landmark \$4bn+/Q level. **Infineon** echoed similar margin disappointment, with c.23% Q4 segment-margin guidance falling short of the mid-20s% buy-side expectation, as mix and FX offset the benefits of stronger AI and cyclical volumes. Conversely, **Renesas** stands out as a margin winner this Q, driven primarily by surging AI datacenter demand, higher utilisation and better industrials. **Overall, we think margin expectations had overrun the attainable for Q4, especially with some of our European names, but that end market drivers remain strong.**

...but cycle fundamentals are robust into 2H26. Largely as expected, commentary this quarter pointed to a demand-led up-cycle rather than a short-lived inventory spike, with industrial strength broadening and automotive beginning to join the recovery. **ON, Infineon** and **Allegro** all noted improving industrial demand alongside steady-to-strengthening automotive, while **TI** saw automotive inflect sharply and **NXP** expects every end market and region to grow sequentially. Channel signals are also constructive: **STM's** distributor inventory has fallen below its normal target, **Allegro's** final pocket of excess distribution inventory has cleared and **NXP's** Western Tier 1s are still ordering hand-to-mouth, suggesting current growth is not simply being pulled forward by aggressive restocking. Pricing is becoming supportive, with sequential increases across several of our global coverage, although the benefit remains selective and, for now, secondary to volume. Visibility is also improving materially, with book-to-bills above 1 and backlog extending through 2026. **Taken together, cleaner channels, firmer pricing and longer-dated order visibility give us greater confidence that the recovery is broadening beyond data center and can carry into 2027.**

2026-27 AI datacenter guidance firms in power semis. Datacenter sales remain the structural overlay, with **ON, TXN** and **Infineon** raising their 2026 outlooks and **STM** lifting its 2027 ambition to well above \$2bn. **TI's** AI DC business has been a standout growth driver, with revenue doubling yy in 2Q26 and management guiding to outgrow the market into 2027 as ASSP design-ins ramp. Subsequently, AI DC sales guidance for FY26 and FY27 was raised c.10% and group GM% lifted (2-3pps), partly reflecting rising datacenter exposure and better factory loading. **ON Semi** sales are also guided to more than double in 2026 (2025: >\$250m), driven by power content wins, with customers ordering into 2027-2028 to lock up supply, supporting greater forward visibility. Elsewhere, **Infineon's** AI datacenter revenue is on track to exceed €1.6bn in FY26 and is poised for a material upgrade to its €2.5bn FY27 target, driven by broad power content wins across the rack, supply-constrained demand, and multi-year customer reservation agreements (CRAs) underpinning strong forward visibility. All told, **power semis sales in the AI DC have gone materially higher of late on volume uptake (regardless of 800v delays or otherwise).** And yet, we guard against runaway assumptions, especially into FY28, given the lack of visibility currently.

What next? For data center power updates we now look to the **AI Infra Summit** in September and the **OCP Global Summit** in October as opportunities for incremental commentary on the timeline for 800v rack deployments and adoption of power semi components. For networking, we think the **Silicon Photonics Global Summit**

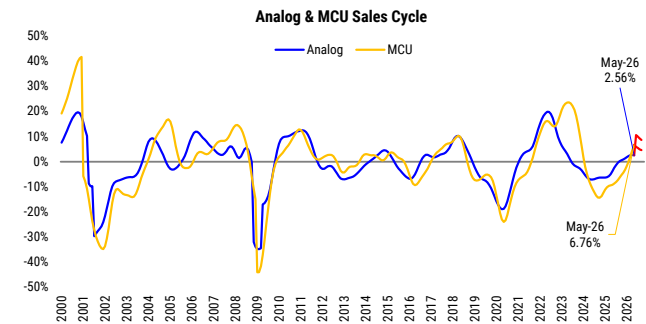
MORGAN STANLEY MUFG SECURITIES CO., LTD.+
Hidetaka Suzuki
Research Associate
Hidetaka.Suzuki@morganstanleymufg.com +81 3 6836-8402

TECHNOLOGY - EUROPEAN SEMICONDUCTORS	
Europe	
Industry View	In-Line
GREATER CHINA TECHNOLOGY SEMICONDUCTORS	
Asia Pacific	
Industry View	Attractive
JAPAN SEMICONDUCTORS	
Japan	
Industry View	In-Line
SEMICONDUCTORS	
North America	
Industry View	Attractive

next week and the **PIC Summit Europe** in November may provide updates on CPO/NPO timelines and optical technology. At the company level, aside from 3Q results in November, we think **ON's Financial Analyst Day** in September could provide broader analog data points on inventory, pricing and end-markets.

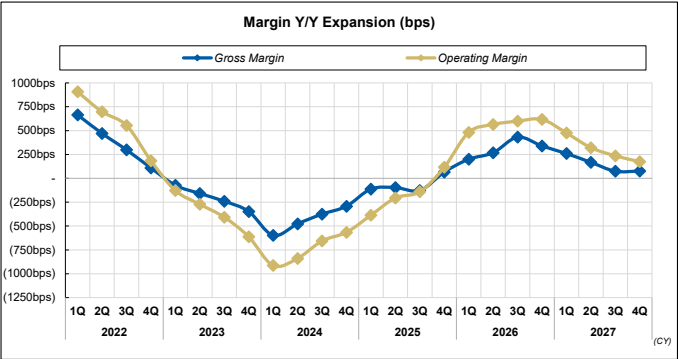
2Q26 in Charts

Exhibit 1: Cycle recovery tracking in line with our expectations...



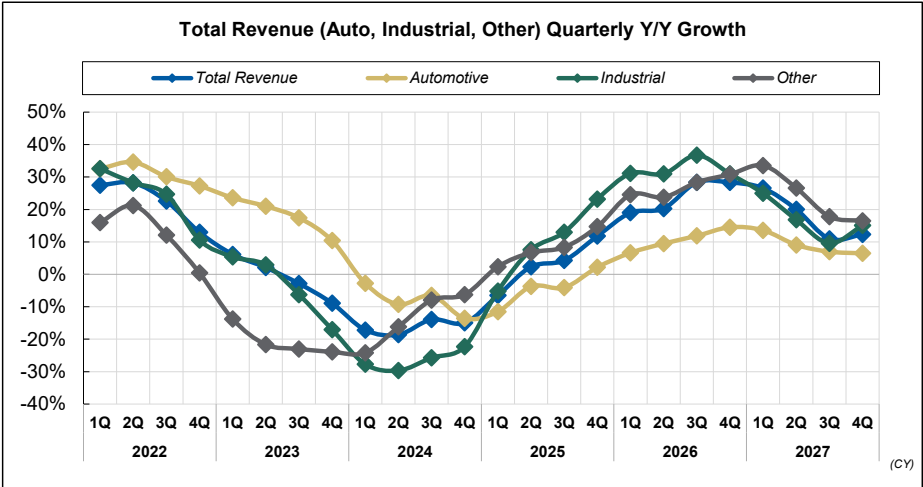
Source: AlphaWise, SIA, Morgan Stanley Research.

Exhibit 2: ...as margins tick sequentially up...



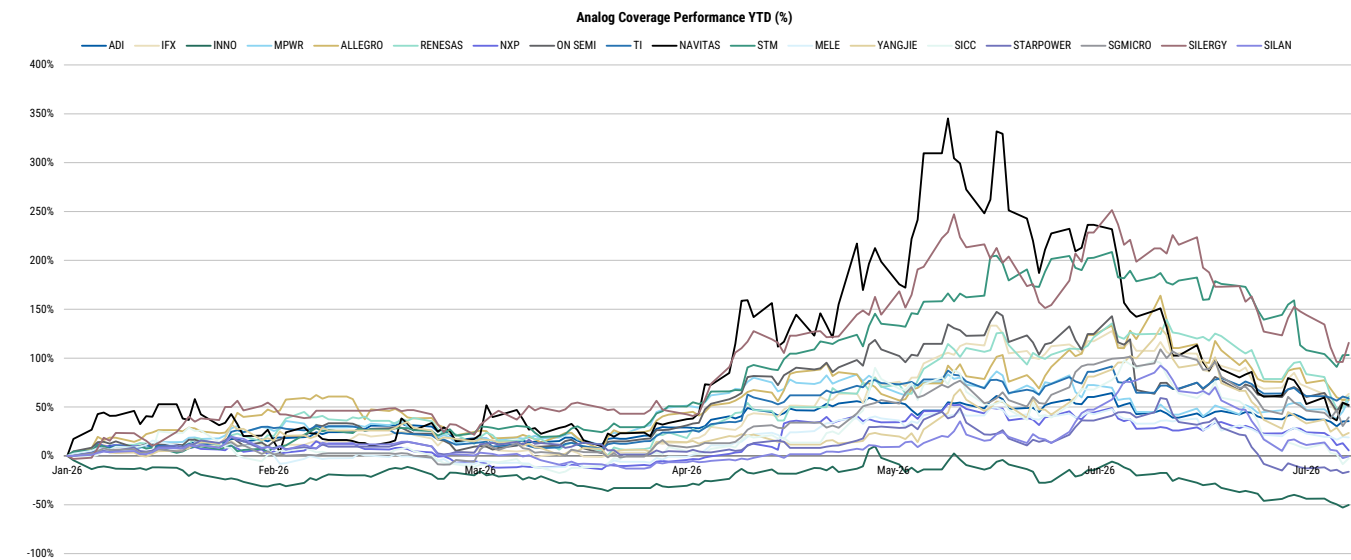
Note: Represents total of defined Core 8 companies (TXN, ADI, IFX, MPWR, STM, NXPI, Renesas, MCHP, ON) Source: VisibleAlpha, Morgan Stanley Research estimates.

Exhibit 3: ...and end-market dynamics accelerate.



Note: Represents total of defined Core 8 companies (TXN, ADI, IFX, MPWR, STM, NXPI, Renesas, MCHP, ON) Source: VisibleAlpha, Morgan Stanley Research estimates.

Exhibit 4: Our Analog covered companies have traded down from May/June peaks



Source: Factset, Morgan Stanley Research estimates

Exhibit 5: Our global analog coverage OW rated stocks

Stock	Rating	Thesis
	OW, PT €81 Covered by Lee Simpson	We view Infineon as best placed amongst European analog names for a cyclical recovery in automotive/industrial, demonstrating a superior ability to weather the headwinds facing these markets in the meantime. We expect new wins in autos to validate IFX's strategy (design focus, capacity build, proprietary IP). A strong IP library augurs well for GaN longer term too including use in EVs, data centres and energy storage.
 life.augmented	OW, PT €65 Covered by Lee Simpson	STM has a portfolio positioned for growth areas (BICMOS imaging sensors, GP MCUs). We see opportunity for recovery into FY26/27. We expect drivers of FY27 growth to come from data centres, with a material ramp in optical revenues driven by pluggables and NPOs (2027+), as well as LEOs in the CECF end market. We allow for a modest industrial recovery cycle starting 2H26. We see some signs of automotive recovery as ASIL-grade microcontroller (MCUs) design wins start to convert to sales (2027+).
 AHEAD OF WHAT'S POSSIBLE™	OW, PT \$458 Covered by Joe Moore	We are OW ADI on the back of 3 reasons: 1) ADI has managed the downcycle well, only seeing 1 quarter of OPM% below 40%, 2) the company's revenue outlook looks strong with high industrial exposure, and 3) the company has relatively higher ASP products which should limit direct competition with emerging Chinese capacity. ADI is a core holding in the analog space long term as the company has exposure to the right end markets as well as superior margins and FCF generation, a result of developing a comprehensive analog portfolio via successful integration of Linear Tech and Maxim.
	OW, PT \$338 Covered by Joe Moore	We remain OW NXP on the back of: 1) NXP should outperform Auto semis in 2026 as auto joins other end markets in a recovery. 2) Structural improvements to the business: NXP has multiple levers to improve gross margin from here (channel shipments, internal utilization, and overall revenue growth); and 3) Favorable MCU set up into 2026 as momentum acceleration appears likely to inflect.
 INNOVATION WITH PURPOSE	OW, PT \$52 Covered by Joe Moore	We see scope for Allegro to outperform over the next several years as the recovery remains in an upcycle, with earnings power increasingly supported by incremental secular growth vectors in auto electrification and industrial. 35x is in-line with the company's average PE multiple since 2022, but is a premium to auto semi peers and in-line with higher gross margin analog/MCU companies.
	OW, PT 6,000 JPY Covered by Kazuo Yosikawa	After years of stagnation, we expect growth of memory interface and digital power for datacenters, MCU for E/E architecture, R-Car for ADAS to drive revenue growth from CY26. Product mix improvement should drive continued gross margin improvement. Data center revenue growth and sustained gross margin improvement outlook, as well as improved prospect of shareholder returns, could trigger re-rating.
 YANGJIE TECHNOLOGY	OW, PT 136 CNY Covered by Daisy Dai	Yangjie is a quality company that we expect to outperform relative to its power semi peers. Overseas packaging and foundry capacity expansion is ongoing. We expect its fab in Vietnam to add growth and gain orders from overseas customers. We expect Yangjie's GM to continue to improve as its product selling prices are stabilizing, auto business is increasing in the mix, and operational efficiency continues to improve. We view valuation as attractive.
	OW, PT 127 CNY Covered by Daisy Dai	We expect SICC to gain share over its global and local competitors due to its technology leadership, especially in 8-inch. Substrate pricing is stabilizing this year, and we think SICC's technology breakthrough could lead to lower costs, thus offering some protection in case pricing starts to decline again in 2027. We expect SiC penetration in EV and data centers to continue to rise. We expect SICC to trade up to 8.5x 2026e BVPS, above the historical average level of 5.5x, owing to of global market share gain and long-term secular drivers (EV, AI data centers, AR glass).

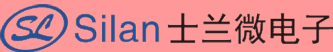
Source: Morgan Stanley Research

Exhibit 6: Our global analog coverage EW rated stocks

Stock	Rating	Thesis
	EW, PT €65 Covered by Nigel van Putten	Melexis enters 2H26 with higher revenue expectations, but growth still trails peers despite a weaker first-half base. Margin conversion remains muted as management prioritises volume over pricing and wage inflation absorbs gross margin gains, while robotics remains too small to affect the near-term earnings path. We remain EW pending clearer evidence of sustained catch-up growth and stronger operating leverage.
	EW, PT \$103 Covered by Joe Moore	There's opportunity here as we think certain concerns surrounding the company are overblown (GM%, BS), but the biggest variable is revenue growth as a cyclical recovery would be an undeniable lever to EPS as well as dispel BS worries. There are considerable headwinds the company will have to get past including increased MCU competition, macro uncertainty, and regaining customer trust from overly extending PSP. If macro uncertainty subsides and we get indication that the MCU upcycle is strengthening, we would be quick to revisit our EW rating, but for now we remain EW.
	EW, PT 69 HKD Covered by Daisy Dai	We believe Innoscience is well positioned to benefit from secular growth drivers for the GaN market, such as NVIDIA's 800V AI data centers, EVs and humanoid robotics. Despite these secular growth drivers, we expect competition to remain fierce and excessive capacity may persist industry-wide in the next five years on the back of aggressive capacity additions. We expect Innoscience's gross margin to improve at a moderate pace in 2026/27e, partly because of rising depreciation costs, and the company to incur losses until 2026e. Our EW relative to our coverage reflects balanced risk-reward on high revenue growth potential, amid lofty market expectations.
	EW, PT 82 CNY Covered by Charlie Chan	The analog cycle has bottomed, but the rebound seems soft in sectors other than auto and AI. SG Micro should still enjoy top-line growth, backed by its comprehensive product portfolio (>6,800 SKUs). The long-term gross margin outlook should be stable, ranging from 48% to 52%, due to improvement in product mix and aggressive market share gains. SG Micro is actively engaging into optical networking business with ability to supply 800G/1.6T optical module. The stock is trading at 65x 2026e P/E, which is less compelling to us as it stands around its historical average.
	EW, PT 120 CNY Covered by Daisy Dai	IGBT is in cyclical recovery, but weaker EV sales this year create headwinds. StarPower can drive revenue growth through IPM and SiC MOS, although the auto MCU development cycle is long, so we expect limited near-term contribution. Margins may remain structurally lower due to the depreciation burden. Our price target implies 45x 2027e P/E (vs. a historical range of 27-100x), which we consider fair.

Source: Morgan Stanley Research

Exhibit 7: Our global analog coverage UW rated stocks

Stock	Rating	Thesis
	UW, PT \$255 Covered by Joe Moore	TI is looking to ramp capex over the next few years to help bring more manufacturing in-house, but this will act as a near-term headwind to earnings and FCF growth. The company has also seen pricing and market share pressure in personal electronics and enterprise, as customers move to multisource components, and face competition with local Chinese supply. The company trades at a premium valuation for its solid governance track record, and we admire the company's long-term direction, but we see this dynamic as limiting upside in the near-term as margin should come under pressure from the capex ramp.
	UW, PT \$12.6 Covered by Joe Moore	The GaN market is increasingly looking more competitive, with STMicro, Infineon, and Renesas all making inroads into GaN via acquisitions and partnerships. While these larger players may expedite the adoption of GaN in markets outside smartphone chargers and expand the TAM, we expect these opportunities to be captured by more established power semi providers rather than Navitas given their supply assurance and automotive qualification experience. NVDA's 800V AI server racks are an enticing opportunity, but given competition and timeline, we think it is too early to fully price in this opportunity.
	UW, PT 388 TWD Covered by Charlie Chan	We have downgraded Silergy in view of its margin miss from foundry cost hikes and weak consumer demand. Its server and optical PMIC exposure remains relatively small. Our price target implies 23x 2027e P/E. We view valuation as more than full, especially after a strong stock rally.
	UW, PT 26.9 CNY Covered by Daisy Dai	We view Silan as a key beneficiary of China's power semi localization trend, but interest expense and depreciation costs weigh on earnings. We expect operating margin of around 10% for 2026, with limited operating leverage benefit despite full utilization. On our estimates, valuation looks demanding at 4.8x 2026 BVPS, vs. the past three year historical average of 3.3x.

Source: Morgan Stanley Research

Exhibit 8: Catalyst Calendar - What's Next?

August	
28.08.26	Innoscence 1H26 Earnings
28.08.26	SG Micro 2Q26 Earnings
31.08.26 - 01.09.26	Silicon Photonics Global Summit
September	
02.09.26-04.09.26	SEMICON Taiwan
04.09.26	Silergy 2Q26 Earnings
15.09.26-17.09.26	AI Infra Summit 2026
16.09.26	ON Financial Analyst Day
October	
12.10.26-15.10.26	OCP Global Summit
13.10.26-15.10.26	SEMICON West
20.10.26-22.10.26	GTC Berlin
19.10.26	Texas Instruments 3Q26 Earnings
27.10.26	NXP 3Q26 Earnings
28.10.26	Melexis 3Q26 Earnings
28.10.26	SICC 3Q26 Earnings
29.10.26	STMicroelectronics 3Q26 Earnings
29.10.26	Renesas 3Q26 Earnings
29.10.26	Allegro Microsystems 2Q27 Earnings
November	
02.11.26	ON 3Q26 Earnings
02.11.26	Navitas 3Q26 Earnings
03.11.26-04.11.26	PIC Summit Europe
05.11.26	Microchip 2Q27 Earnings
10.11.26	Infineon 4Q26 Earnings
24.11.26	Analog Devices 4Q26 Earnings

Source: Morgan Stanley Research

Where are we in the cycle

Broadening Upcycle

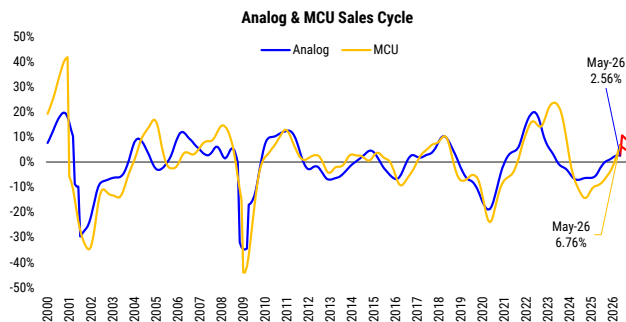
Commentary this quarter aligned with our expectations: **improving end market dynamics supported by stabilising prices and lean channel inventories**. Commentary supports signs of an upcycle with the backlog presenting the strongest signal as book to bills exceed one. While pricing is not yet broad-based, there is uniform dialogue across our coverage of at least near-neutral prices. ON Semi, Infineon and STMicro are all implementing selective price increases to offset costs and address tight supply. This confirms our thesis that prices have troughed and sequential price increases will likely continue for the rest of the year.

Channel conditions are uniformly healthier: distributor and customer inventories are at, below or near normal levels. However, the absence of broad restocking argues that current growth is demand-led rather than an inventory pull-forward. IFX is furthest into replenishment; TXN and MPWR report particularly lean positions, while ON, NXPI and ALGM still see cautious ordering. Hand-to-mouth ordering behaviour was observed at Western Tier 1 auto OEMs, which we think is the clearest signal that the destocking cycle has ended and a replenishment wave is building.

End-market dynamics picking up. Most of our coverage guided to above seasonal end-market growth reaffirming positive indicators from our Distributor Survey ([here](#)). Looking at autos, drivers were mixed across our coverage with commentary of restocking, China EV tailwinds and SDV content gains. Industrial dynamics were also signalled as improving. STMicroelectronics lifted industrial guidance to 40% yy improvement, vs prior guide of 30% yy for Q4, with management speaking to a broader recovery benefitting sales of GP MCUs.

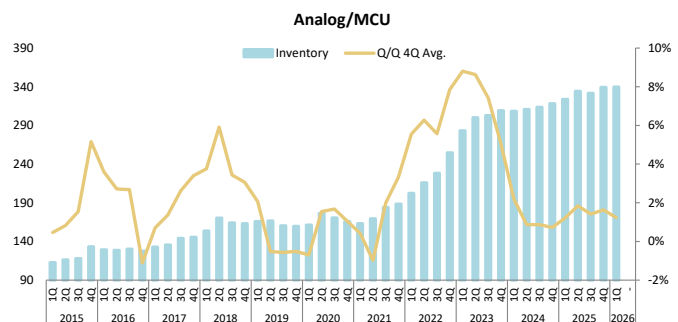
What does this mean for 2H26? In our view, this quarter has demonstrated strong signals of the start of an up-cycle. We expect this to continue for at least the latter half of this year, with likely continued pricing improvements, steps towards inventory replenishment and constructive book-to-bills. We expect sequential selective price increases for at least the rest of this year, with this likely becoming more broad-based into 2H26. We also expect more of our global coverage to start talking to book-to-bills exceeding one as channel inventories remain relatively lean and end-market activity continues to pick up.

Exhibit 9: Analog + MCU are now shipping to end demand/ slightly above as of May



Source: AlphaWise, SIA, Morgan Stanley Research.

Exhibit 10: Inventories remain historically elevated vs history, but momentum has stabilized



Source: Factset, Morgan Stanley Research.

Stocks

Europe Semiconductors (Lee Simpson, Nigel van Putten, Amelia Scicluna)

There was a lot to like this quarter for **Infineon** (IFX, OW, PT €81): data center guidance was raised and automotive and industrial growth dynamics were reaffirmed. Yet the stock reaction on the day of the print only tells the tale of the segment margin miss. Although the company beat our preview of a 22% Q4 segment margin guide, the 23% guide was insufficient to appease the bulls hoping for closer to 24/25%. We think the sell-off misses the bigger picture. While we see IFX's Q4 segment margin as unlikely to hit 25%, we think growing utilisation, improving pricing and a changing product mix will allow segment margins to hit 25% in FY27. We remain OW Infineon applying a SOTP valuation framework to reflect the differing growth trajectories of the data center and cycle segments. We view the recent sell-off as overdone and think the structural breakout story remains underappreciated.

For **STMicroelectronics** (STM, OW, PT €65), management communication once again weighed on its share price this quarter. Like with Infineon, data center guidance was raised, however margin missteps sent the stock down c.18% on earnings day. The company previously guided to attaining a 40% GM% once \$4bn sales were hit; however, on the conference call management walked back from this, citing two key headwinds: elevated unloading charges from China and ongoing costs from technology transfer to facilitate the manufacturing reshaping programme. Now the dust has settled on the print, we think the LEO and data center segments remain underappreciated growth drivers for the stock. We remain OW STMicroelectronics and value the stock on a SOTP basis to reflect the different growth profiles of the LEO, Data Center and cyclical segments.

We raised estimates for **Melexis** (MELE, EW, PT €65) post print reflecting a 2Q beat, a higher second-half guidance and a better gross margin path. However, the revisions also reflect a catch-up to a broader sector recovery that became clearer during the quarter. However, the second-half revenue guide still underwhelms as it implies c.8% growth versus the first half, below c.10% for peers and from a weaker first-half base. Margin conversion remains underwhelming, with EBIT margin guided to around 18% and the gross margin, although improving, may not return to the levels previously seen (>43%) given

increased competition from China.

North America Semiconductors (Joe Moore, Nicole Kozhukhov)

Analog: Recovery broadening beyond AI into industrial and auto. **TXN** beat across the board - Industrial up ~30% y/y, Auto at +mid-teens y/y, and DC doubling - with Q3 guided well above Street; customers remain early in the recovery and not yet building inventory, a constructive 2H setup, though we stay UW on valuation. **Allegro** reinforced the theme, with DC reaching a record 17% of sales and FY27 DC guided to more than double y/y, while Auto grew ahead of its >10% model target. **ADI** reinforced the recovery and saw another meaningful beat and raise - with the OctQ guide implying the 10th consecutive above-seasonal quarter, B/B remaining >1 across DC, Industrial, and Consumer with no inventory build, and DC growing >100% y/y in both power and optical. What stood out the most is that mgmt also effectively raised the LT growth framework from +MSD% toward double digits for the next several years. We reiterate OW, viewing recent underperformance as a significant buying opportunity.

Power Semi: Demand recovery remains intact, with lead times extending and customers ordering into 2027-2028 to secure supply. Strengthening demand has created some near-term constraints in certain power products, with **ON** prioritizing AI data center shipments over automotive and industrial in 2Q while manufacturing catches up to demand over the coming quarters. In auto, ON now expects China SiC revenue to grow 60-70% y/y in 2026, supported by continued share gains and ramps across existing and new vehicle programs, while ON also expects ESS revenue to grow ~40% y/y, driven by North American microgrid customers.

MCU: Strong results across the group, with DC the standout driver. **MCHP** had a very strong quarter – inventory correction firmly behind it, DC revenue up ~98% y/y in JunQ and total DC exposure guided to ~\$1bn in CY26 (~69% y/y), with PCIe Gen6 switch design wins doubling to 12 programs (in addition to 2 Gen6 retimer and parts now in early production. **NXP** was in line, though auto was up +17% y/y, driven entirely by content gains from SDV architecture transitions with no restocking tailwind, and the Kinara Physical AI pipeline growing to \$1.5bn+ (from \$1bn+ last quarter) across 200+ unique customers, though revenue conversion remains a 2027+ event. DC emerging as a third growth engine, with revenue expected to exceed \$500mn in 2026 vs. ~\$200mn in 2025.

Greater China Semiconductors (Charlie Chan, Daisy Dai, Henry Zhao, Lucas Wang)

Power semi: improving demand, but uneven earnings conversion across companies.

Yangjie showed the strongest momentum, supported by robust automotive demand, and healthy demand across AI, energy storage and industrial applications, suggesting the broader demand recovery remains intact. In contrast, StarPower and Silan Micro both saw earnings disappoint despite improving industry pricing: StarPower continued to face higher input costs and a lag in passing through price increases, while Silan's earlier price hikes had yet to translate into meaningful margin improvement.

SIC/GaN: SICC benefiting from volume, Innoscience awaiting data center inflection: We expect China SiC device companies to continue to gain market share in the global SiC market, for instance UNT (688469.SS, NC) guided 2026 SiC device revenue to grow

~100% and to reach Rmb3.0bn, ~70-80% from volume growth and the rest from ASP hikes, outgrowing global peers. Also they will expand SiC 8-inch capacity from 7k wpm now to 15k wpm by end of 2026, so SICC will be the beneficiary as one of SiC substrate suppliers. As for GaN, the inflection for GaN in data center seems to be in 2028 at the earliest, which bodes ill for InnoScience, although we are still confident regarding 65% revenue growth in 2026, mainly driven by industrial applications.

MCU: chip inflation to sustain in 2H26. The Greater China MCU market remains in a supply-driven up-cycle, with tight mature-node capacity and rising foundry costs supporting continued chip price increases into 2H26. Demand is mixed – strong in data-center applications, stable in industrial and weak in consumer – while declining inventory supports a healthier supply-demand balance.

China Analog: Recovery broadening, but remains structurally bifurcated. We are turning incrementally more constructive on China analog. Demand recovery appears to be broadening beyond consumer, with industrial/energy and AI-related networking showing stronger growth, while inventory and lead times remain at 8-10 weeks. More importantly, pricing pressure could be easing as global analog vendors become more disciplined on pricing, although we do not see evidence of an industry-wide price upcycle for China analog yet. We believe AI infrastructure, particularly optical modules and servers, is emerging as an incremental growth driver, while continued consolidation among domestic vendors could benefit leading platform players over time.

SG Micro (300671.SZ, EW): We see increasing upside optionality from AI-related optical modules. Company's networking/computing is among its fastest-growing end markets, with optical module revenue growing strongly. We see rising analog content as the industry transitions from 800G to 1.6T, driven by additional channels and higher-value signal-chain products, which could support both revenue growth and mix improvement. That said, we remain EW as broader China analog demand recovery remains uneven and valuation needs to be balanced against still-elevated R&D investment.

Silergy (6415.TW, UW): We believe Silergy will also benefit from the recent cyclical tailwind. China industrial and automotive customers have started to replenish inventory after a muted 12-month period, while AI and memory also saw rush orders over the past quarter. However, with rising production cost, we expect Silergy to pass through only part of the increase to customers, and migration to 12-inch will take time. We therefore see near-term margin pressure, as more than 80% of revenue still comes from 8-inch, with most of the impact likely in 2Q26–3Q26 as wafer prices rise.

Japan Semiconductors (Kazuo Yoshikawa, Hidetaka Suzuki)

Renesas Electronics (6758.T, OW): AI Strength Spilling Over into Adjacent Markets: Renesas delivered a positive earnings surprise in 2Q (Jun-Q). In Automotive, the R-Car Gen 4 ramp progressed smoothly with Japanese customers. In Industrial, strong demand from factory automation and China supported growth during the quarter. Datacenter continued to grow, led by digital power products for ASICs, while memory interface products also maintained solid momentum. We also viewed Renesas' selective price increases from July as a positive, although management said that this is an extension of its existing pricing strategy and is not expected to drive an immediate increase in gross margins.

We had thought that, in addition to secular AI/datacenter growth, cyclical recovery of

semiconductor market would start in a wider range of end markets. However, CEO Shibata's comment that industrial strength is being driven by datacenter demand suggests a different dynamic: Secular growth of AI/datacenter is now spilling over into adjacent industries such as factory automation and semi cap equipment.

2Q26 Surprises

Incremental Data Center Updates

Broad-based data center guidance upside, but stocks only selectively rewarded. Into this quarter's prints, eyes were on 800v data center commentary after Jensen Huang said that Nvidia was replacing the Kyber rack architecture with a new solution. Despite some bears expecting this to lower power semi content, those in our coverage exposed to the power semi theme all either reiterated or raised their data center guidance for FY26 and FY27. ON, NVTs, TI and IFX all reiterated that the transition to 800v still presents a step-up in power semi content from the existing architecture. Importantly, while IFX clarified that the \$/kW doesn't change materially in the transition to 800v, the \$/rack is incrementally larger. On networking, STM confirmed optic strength citing broadening customer partnerships, with management expecting PIC/EIC sales to pick up materially next year. Despite this incrementally positive commentary, share prices weren't rewarded on the earnings day, with most trading off on margin concerns.

Powering the Data Center

Infineon (OW, PT €81) Data center guidance raised and CRAs announced. Like many bulls were hoping for, Infineon raised its data center guidance marginally for FY26 to €1.6bn (prior €1.5bn) and hinted at meaningful upside to the €2.5bn guide for FY27. Focus now shifts to 4Q26, when many will look for this to be quantified. Interestingly, Infineon spoke to introducing Customer Reservation Agreements (CRAs). These were confirmed to be of \$HSDbn value with a 3-4 year time horizon. Management didn't explicitly confirm the use of prepayments, but these were signalled as being negotiated. We view this as confirmation of our structural breakout story thesis of Infineon as the preferred partner for power semi deployment to the data center. For instance, we note that this scale of customer partnerships hasn't been spoken to by other analog companies.

ON Semiconductor: Data center guidance raised for CY26 and new design win. Data center guidance was raised for CY26, with ON now expecting AIDC to more than double y/y versus doubling previously. While mgmt did not provide detailed 2Q AIDC growth detail, the segment drove the majority of "Other" growth (+34% q/q), with "Other" guided to high-teens q/q growth in 3Q, again led by AIDC. Design-win momentum also broadened in 2Q, including AWS BBU & PSU support, expanded Nvidia MGX ecosystem support, and continued engagements across all leading hyperscalers. ON expects SiC revenue in AIDC to grow ~60% y/y in 2026. Longer-term, ON raised its AIDC SAM to ~\$50bn by 2030 from ~\$12bn previously and sees high-voltage content per rack increasing ~10x, and it expects the transition to 800V DC to begin ramping in late 2027/early 2028.

ADI: Empower opportunity in early stages, but high long-term potential. Following the July close of its \$1.5bn Empower acquisition, ADI provided incremental color on its contribution to DC growth, with Empower expected to add several points in FY27, primarily in power, and a larger IVR opportunity beginning in 2028+. ADI also noted that its 2030 data center and energy SAM has more than doubled versus what it envisioned a year ago, with mgmt framing the analog opportunity at ~\$1-1.5bn per GW of data center infrastructure.

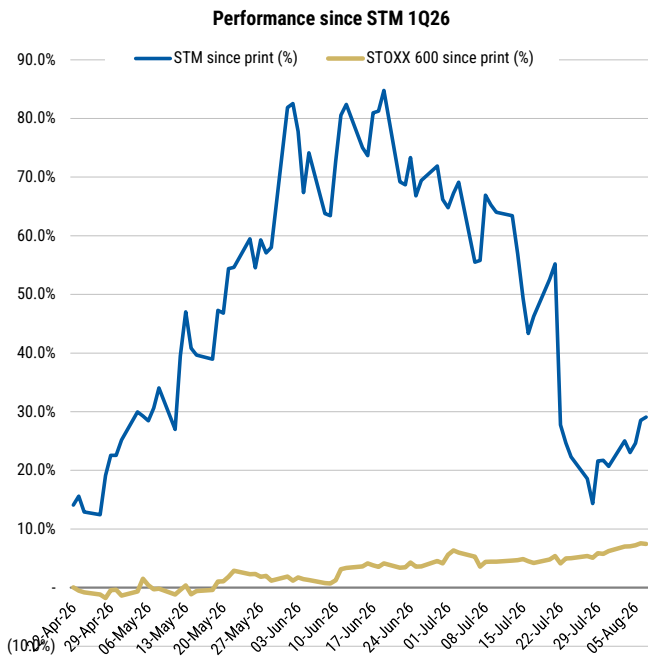
Optical Networking

STMicroelectronics (OW, PT €65) Building capacity to meet rising demand. While STMicro traded down on the day of results, there was a lot to like from its data center commentary. Management raised data center guidance for the 3rd time this year to more than \$1bn in 2026 and well above \$2bn in 2027, citing strong demand for optical connectivity. Currently our understanding is that the majority of optic sales are from microcontrollers, while next year we expect the main drivers to be from PIC and EIC sales as adoption of 800Gbps and 1.6Tbps pluggable optics accelerates. In our view, STMicro's capacity expansion is strategically positioned to support this rising demand. Namely, the Crolles facility, which management expects to ramp over the next 12 to 18 months. While we appreciate margins missed market expectations, we don't think the share price reflects this material data center opportunity.

Notable Mover: STMicroelectronics

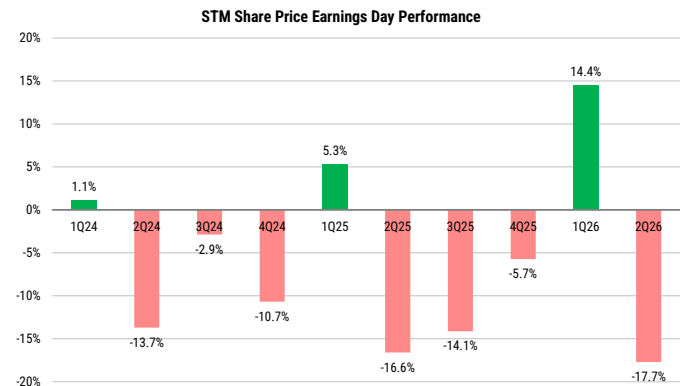
Despite the recent knock to STMicro's share price, the stock still outperforms the STOXX 600 and we expect this to continue given clearly improving end-market dynamics, better communication from the management team on the NDR and the data center guidance uptick. We remain Overweight STMicro but took stock of disappointment in margin leverage by trimming SOTP multiples to 30x for the data center and LEO segments and 15x for the cyclical segment, setting our PT at €65 (see [Margin Leverage in Question](#)).

Exhibit 11: STM daily moves vs STOXX 600 Europe since the Q1 print



Source: FactSet, Morgan Stanley Research

Exhibit 12: STMPA Earnings Day Performance since 1Q24

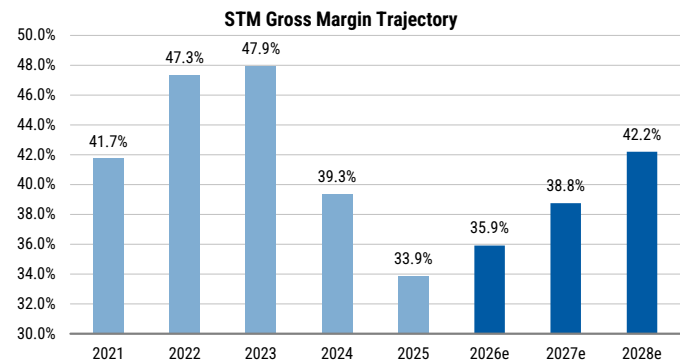


Source: FactSet, Morgan Stanley Research

Margin disappointment. Much of investor disappointment at the print came from

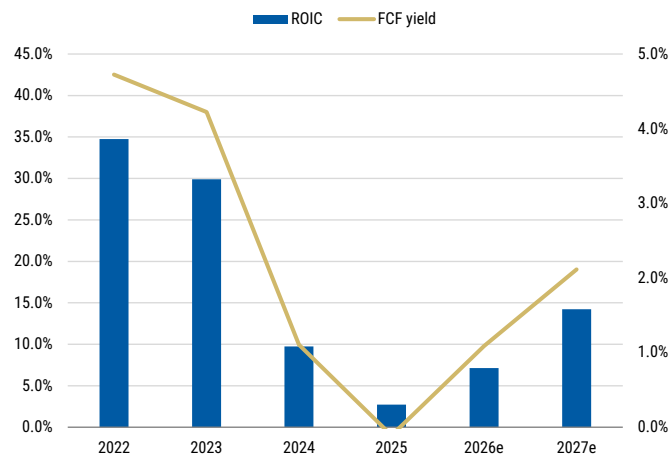
management highlighting that gross margins won't reach 40% in Q4. Management detailed the moving parts behind this, calling out technology transfer costs as a 50-60bps headwind and persistent underloading charges (due notably to China) representing a 70bps headwind. However, key for longer-term margins is the development of the restructuring programme as capacity increases and transition to larger wafer diameters supports margins. There's also the chance that pricing dynamics are improving and the company has signalled possible ongoing sequential price improvements through the rest of this year. However, this isn't uniform across all segments and customers, for instance pricing difficulties persist in the China automotive segment.

Exhibit 13: We expect 200bps yy GM% improvement FY25 to FY26 before expanding by 290bps to FY27



Source: Company data, Morgan Stanley Research estimates

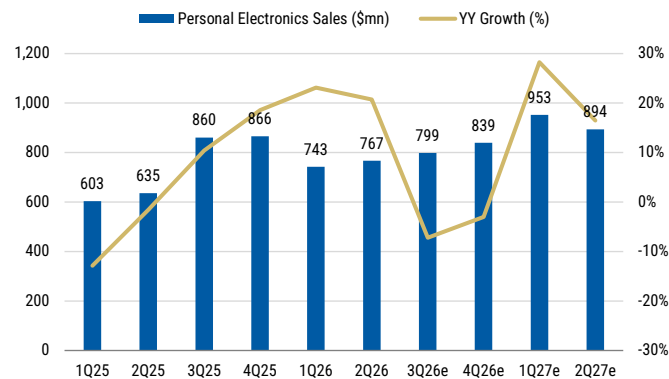
Exhibit 14: Valuation metrics signal 2025 as the trough



Source: Company data, Morgan Stanley Research estimates

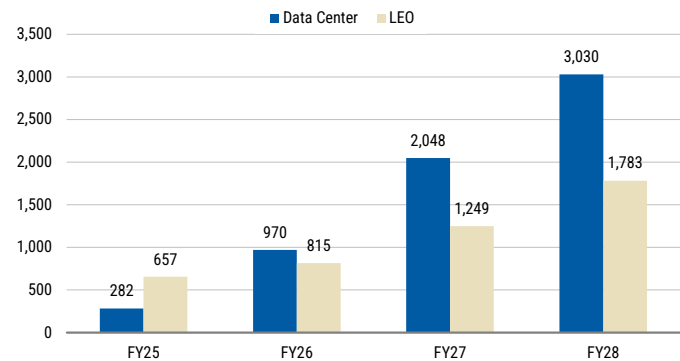
A transition year for personal electronics. Management circled in on the personal electronics (PE) segment as the weakest revenue driver this year, expecting a low- to mid-single-digit increase yy (but with a YY decline in H2). The changing demand profile is unsurprising given memory headwinds for handset builds and the change in product pipeline from their main customer, Apple. However, a bright spot in this area is that management called out continued increase in content in the medium term, improving visibility suggesting this is a temporary headwind, not a structural issue. As a reminder, we currently model for PE sales declining 6-7% yy in Q3.

Exhibit 15: We see Personal Electronics sales troughing in 2H26 before recovering in early 27.



Source: Company data, Morgan Stanley Research estimates

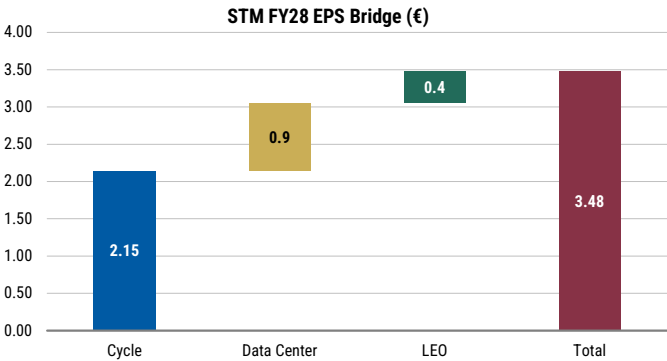
Exhibit 16: DC and LEO annual sales



Source: Company data, Morgan Stanley Research estimates

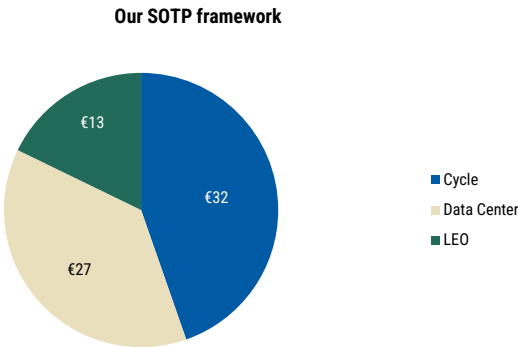
Data center guidance uptick. At the print management raised data centre guidance to 'well above \$2bn' for FY27. Our understanding is that this is largely from silicon photonics and optical strength, with management speaking in more detail about significant STM32 content in 800G and above optical applications. Management reaffirmed the diversified customer base but stressed also the very strong growth expected into year-end with Q4 sales for CECP seen up 80-90% yy.

Exhibit 17: We see STMicro's FY28 EPS driven by the cycle, data center and LEO sales...



Source: Morgan Stanley Research estimates

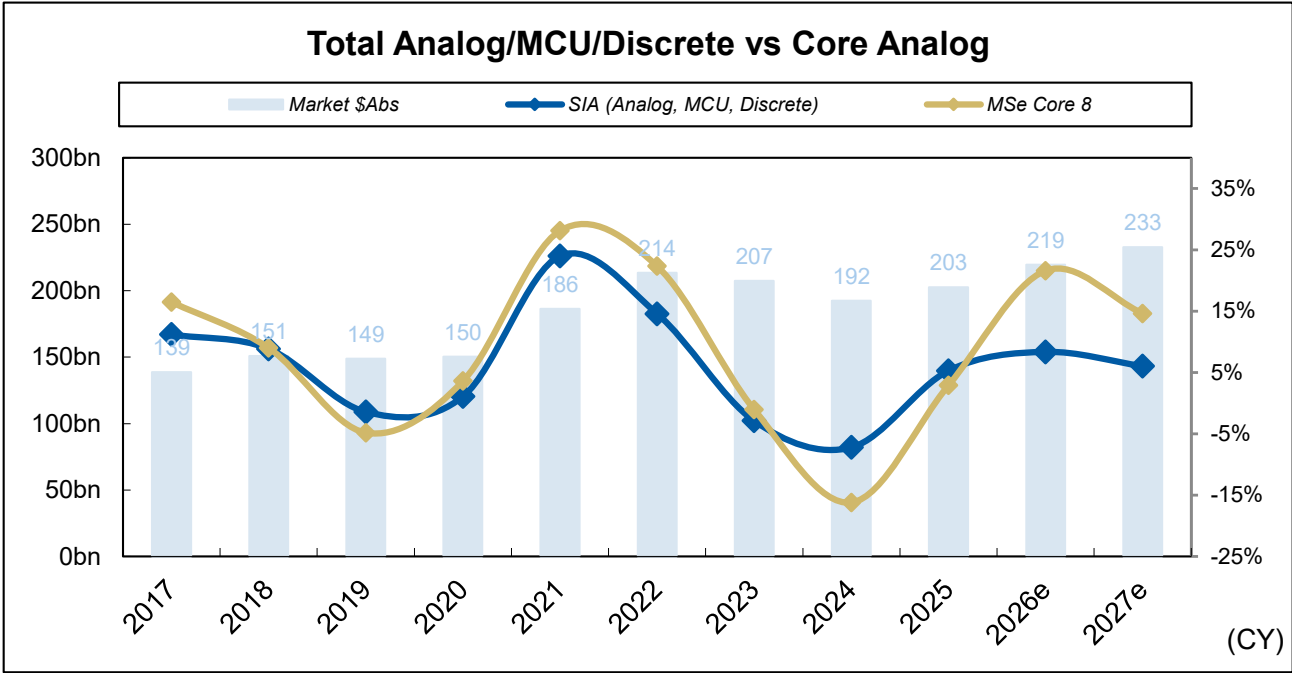
Exhibit 18: ...we use this to derive our SOTP valuation framework.



Source: Morgan Stanley Research estimates

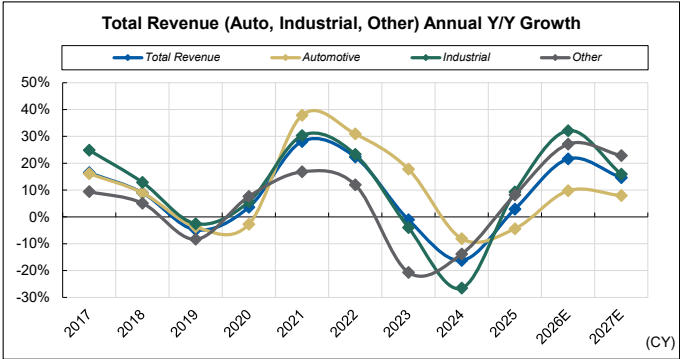
Global Analog Model

Exhibit 19: We model for 8%/6% y/y growth in 2026/2027, while the Core 8 outgrow the market at 22%/15%...



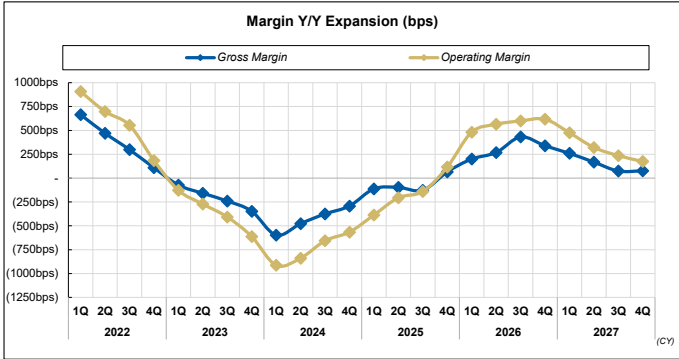
Note: Represents total of defined Core 8 companies (TXN, ADI, IFX, MPWR, STM, NXPI, Renesas, MCHP, ON) Source: VisibleAlpha, SIA, Morgan Stanley Research estimates.

Exhibit 20: ...with all end markets expected to grow in 2026 and 2027...



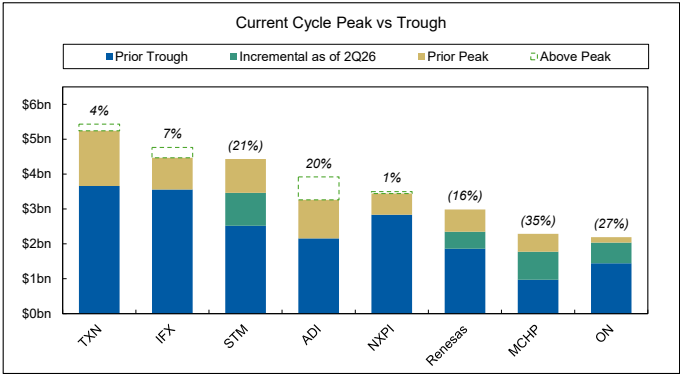
Note: Represents total of defined Core 8 companies (TXN, ADI, IFX, MPWR, STM, NXPI, Renesas, MCHP, ON) Source: VisibleAlpha, Morgan Stanley Research estimates.

Exhibit 21: ...along with continued margin expansion through 2027...



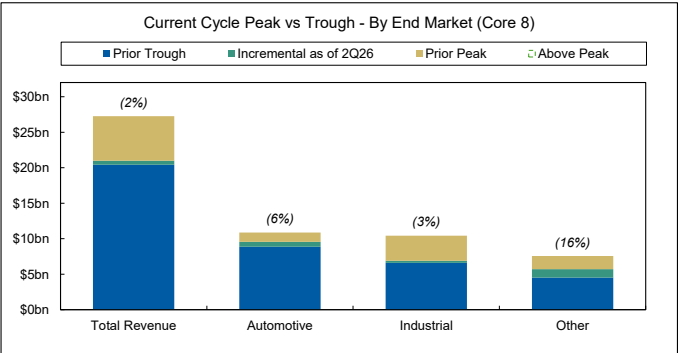
Note: Represents total of defined Core 8 companies (TXN, ADI, IFX, MPWR, STM, NXPI, Renesas, MCHP, ON) Source: VisibleAlpha, Morgan Stanley Research estimates.

Exhibit 22: ...with several companies already above prior-cycle peaks...



Source: VisibleAlpha, Morgan Stanley Research estimates.

Exhibit 23: ...even as aggregate end markets remain below prior-cycle peaks



Source: VisibleAlpha, Morgan Stanley Research estimates.

Exhibit 24:

Global Analog Tracker Revenue Dynamics by Stock

	CY2016	CY2017	CY2018	CY2019	CY2020	CY2021	CY2022	CY2023	CY2024	CY2025	CY2026e	CY2027e	CY2028e
Revenue													
Infinion (€mn)	6,563	7,193	7,794	7,976	9,281	11,588	15,010	16,059	14,677	14,900	16,986	19,422	
STMicroelectronics (\$mn)	6,972	8,346	9,665	9,556	10,219	12,761	16,128	17,286	13,269	11,799	14,489	17,866	19,483
Melexis (€mn)	456	512	569	487	508	644	836	964	933	840	872	933	997
NXP Semiconductor (\$mn)	9,498	9,256	9,407	8,877	8,612	11,063	13,205	13,276	12,614	12,269	14,255	15,767	16,611
Analog Devices (\$mn)	3,636	5,642	6,223	5,754	5,858	8,444	12,579	11,569	9,338	11,757	16,281	18,513	20,577
Texas Instruments (\$mn)	13,370	14,961	15,784	14,383	14,461	18,344	20,028	17,519	15,641	17,682	22,018	23,832	25,137
Microchip (\$mn)	3,063	3,881	5,022	5,278	5,298	6,444	8,050	8,541	4,757	4,372	6,024	7,095	7,800
Navitas Semiconductor (\$mn)	-	-	-	-	12	24	38	79	83	46	48	68	98
ON Semiconductor (\$mn)	3,907	5,543	5,878	5,518	5,255	6,740	8,326	8,253	7,082	5,995	6,566	7,551	8,654
SG Micro (RMB mn)	452	532	572	792	1,197	2,238	3,188	2,616	3,347	3,898	4,961	5,852	6,744
Silergy (NT\$ mn)	7,139	8,599	9,414	10,778	13,876	21,506	23,511	15,427	18,455	18,812	23,695	28,515	33,920
Renesas (¥bn)	639	779	757	718	716	994	1,503	1,469	1,348	1,318	1,638	1,875	2,093
Innoscence (RMB mn)						68	136	593	828	1,213	2,001	3,641	6,176
Yangjie Technology (RMB mn)	1,190	1,470	1,852	2,007	2,617	4,397	5,403	5,410	6,033	7,130	9,433	11,517	14,182
SICC (RMB mn)			136	269	425	494	417	1,251	1,768	1,465	2,231	2,817	3,123
StarPower (RMB mn)	300	437	671	779	963	1,707	2,705	3,663	3,391	4,012	4,850	5,799	7,318
Allegro MicroSystems (\$mn)					591	743	905	1,078	773	840	1,060	1,264	1,535
Silan Microelectronics (RMB mn)	2,375	2,742	3,026	3,111	4,281	7,194	8,282	9,340	11,221	13,052	15,805	18,063	20,443
YY(%)													
Infinion		9.6%	8.4%	2.3%	16.4%	24.9%	29.5%	7.0%	-8.6%	1.5%	14.0%	14.3%	
STMicroelectronics		19.7%	15.8%	-1.1%	6.9%	24.9%	26.4%	7.2%	-23.2%	-11.1%	22.8%	23.3%	9.1%
Melexis		12.1%	11.3%	-14.5%	4.3%	26.8%	29.9%	15.3%	-3.3%	-10.0%	3.9%	6.9%	6.9%
NXP Semiconductor		-2.5%	1.6%	-5.6%	-3.0%	28.5%	19.4%	0.5%	-5.0%	-2.7%	16.2%	10.6%	5.4%
Analog Devices		55.1%	10.3%	-7.6%	1.8%	44.1%	49.0%	-8.0%	-19.3%	25.9%	38.5%	13.7%	11.1%
Texas Instruments		11.9%	5.5%	-8.9%	0.5%	26.9%	9.2%	-12.5%	-10.7%	13.0%	24.5%	8.2%	5.5%
Microchip		26.7%	29.4%	5.1%	0.4%	21.6%	24.9%	6.1%	-44.3%	-8.1%	37.8%	17.8%	9.9%
Navitas Semiconductor						100.3%	59.9%	109.4%	4.8%	-44.9%	3.7%	43.3%	43.4%
ON Semiconductor		41.9%	6.0%	-6.1%	-4.8%	28.3%	23.5%	-0.9%	-14.2%	-15.3%	9.5%	15.0%	14.6%
SG Micro		17.6%	7.7%	38.5%	51.0%	87.1%	42.4%	-17.9%	28.0%	16.5%	27.3%	18.0%	15.2%
Silergy		20.5%	9.5%	14.5%	28.8%	55.0%	9.3%	-34.4%	19.6%	1.9%	26.0%	20.3%	19.0%
Renesas		22.0%	-2.9%	-5.1%	-0.4%	38.9%	51.1%	-2.2%	-8.2%	-2.2%	24.3%	14.4%	11.6%
Innoscence							99.6%	335.3%	39.8%	46.4%	64.9%	81.9%	69.6%
Yangjie Technology		23.5%	26.0%	8.4%	30.4%	68.0%	22.9%	0.1%	11.5%	18.2%	32.3%	22.1%	23.1%
SICC				97.3%	58.2%	16.3%	-15.6%	199.9%	41.4%	-17.2%	52.3%	26.3%	10.8%
StarPower		45.6%	53.7%	16.1%	23.6%	77.2%	58.5%	35.4%	-7.4%	18.3%	20.9%	19.6%	26.2%
Allegro MicroSystems						25.9%	21.7%	19.2%	-28.3%	8.7%	26.2%	19.2%	21.5%
Silan Microelectronics		15.4%	10.4%	2.8%	37.6%	68.1%	15.1%	12.8%	20.1%	16.3%	21.1%	14.3%	13.2%
Automotive Segment Revenue													
Infinion (€mn)					3,847	5,081	6,998	8,296	7,559	7,470	7,886	8,678	
STMicroelectronics (\$mn)					-	-	-	7,944	6,064	4,594	5,106	5,237	5,383
Melexis (€mn)	406	455	518	448	447	573	753	868	840	741	785	840	897
NXP Semiconductor (\$mn)	3,688	4,263	4,507	4,212	3,825	5,493	6,879	7,484	7,151	7,116	7,874	8,896	9,582
Analog Devices (\$mn)	1,048	1,471	1,541	1,484	1,453	1,989	2,598	2,975	2,847	3,380	3,946	4,298	4,670
Texas Instruments (\$mn)	2,372	2,846	3,237	3,081	2,896	3,919	4,976	5,919	5,503	5,780	6,406	6,479	6,955
ON Semiconductor (\$mn)	1,321	1,665	1,846	1,819	1,686	2,289	3,361	4,320	3,901	3,081	3,212	3,321	3,505
Silergy (NT\$ mn)						121	962	1,186	1,899	2,280	3,460	4,133	4,916
Renesas (¥bn)	350	413	390	371	341	462	645	660	703	638	740	793	838
Innoscence (RMB mn)								4	42	61	123	246	369
Allegro MicroSystems (\$mn)					385	509	618	773	580	604	680	756	807
YY(%)													
Infinion						32.1%	37.7%	18.5%	-8.9%	-1.2%	5.6%	10.0%	
STMicroelectronics									-23.7%	-24.2%	11.2%	2.6%	2.8%
Melexis		12.1%	13.8%	-13.5%	-0.3%	28.3%	31.4%	15.3%	-3.3%	-11.7%	5.9%	6.9%	6.9%
NXP Semiconductor		15.6%	5.7%	-6.5%	-9.2%	43.6%	25.2%	8.8%	-4.4%	-0.5%	10.6%	13.0%	7.7%
Analog Devices		40.4%	4.7%	-3.7%	-2.1%	36.9%	30.6%	14.5%	-4.3%	18.7%	16.8%	8.9%	8.6%
Texas Instruments													
ON Semiconductor		26.1%	10.9%	-1.5%	-7.3%	35.8%	46.8%	28.5%	-9.7%	-21.0%			
Silergy							697.1%	23.3%	60.1%	20.1%	51.7%	19.5%	19.0%
Renesas		17.8%	-5.6%	-4.8%	-8.1%	35.6%	39.5%	2.4%	6.4%	-9.2%	15.9%	7.2%	5.6%
Innoscence									986.6%	45.1%	100.0%	100.0%	50.0%
Allegro MicroSystems						32.1%	21.5%	25.0%	-25.0%	4.2%	12.5%	11.2%	6.7%
Industrial Segment Revenue													
Infinion (€mn)					1,435	1,562	1,908	2,192	1,787	1,640	1,803	2,142	
STMicroelectronics (\$mn)					-	-	-	3,972	2,657	2,471	3,240	3,891	3,888
NXP Semiconductor (\$mn)	1,471	1,718	1,813	1,599	1,836	2,410	2,713	2,351	2,269	2,273	3,003	3,332	3,516
Analog Devices (\$mn)	2,347	3,582	3,827	3,587	3,884	4,945	6,526	6,015	4,181	5,363	8,048	9,411	10,711
ON Semiconductor (\$mn)	953	1,417	1,594	1,428	1,354	1,800	2,291	2,278	1,801	1,675	1,689	1,833	2,005
Silergy (NT\$ mn)	2,806	3,432	3,646	4,088	4,782	7,288	7,890	5,316	5,875	5,781	7,470	8,924	10,615
Renesas (¥bn)	270	353	347	330	364	516	846	799	637	642	889	1,074	1,248
Innoscence (RMB mn)								228	219	521	893	1,713	2,084
Allegro MicroSystems (\$mn)					205	235	286	305	193	236	380	508	728
YY(%)													
Infinion						8.9%	22.2%	14.9%	-18.5%	-8.2%	10.0%	18.8%	
STMicroelectronics									-33.1%	-7.0%	31.1%	20.1%	-0.1%
NXP Semiconductor		16.8%	5.5%	-11.8%	14.8%	31.3%	12.6%	-13.3%	-3.5%	0.2%	32.1%	10.9%	5.5%
Analog Devices		52.6%	6.8%	-6.3%	8.3%	27.3%	32.0%	-7.8%	-30.5%	28.3%	50.1%	16.9%	13.8%
ON Semiconductor		48.8%	12.5%	-10.4%	-5.2%	32.9%	27.3%	-0.5%	-21.0%	-7.0%	0.9%	8.5%	9.4%
Silergy		22.3%	6.2%	12.1%	17.0%	52.4%	8.3%	-32.6%	10.5%	-1.6%	29.2%	19.5%	19.0%
Renesas		30.8%	-1.9%	-4.9%	10.3%	41.8%	64.1%	-5.5%	-20.3%	0.9%	38.4%	20.8%	16.1%
Innoscence									-4.2%	137.9%	71.6%	91.7%	21.7%
Allegro MicroSystems						14.2%	22.0%	6.8%	-36.8%	22.1%	61.4%	33.6%	43.4%

Source: Company Data, Morgan Stanley Research estimates (e)

Exhibit 25: Global Analog Tracker GM%, FCF, Inventory, ROIC, ROCE by Stock

	CY2016	CY2017	CY2018	CY2019	CY2020	CY2021	CY2022	CY2023	CY2024	CY2025	CY2026e	CY2027e	CY2028e
Gross margin (reported)													
Infineon	36.0%	37.2%	38.7%	36.7%	32.9%	39.6%	44.5%	44.8%	39.9%	39.4%	42.1%	45.6%	
STMicroelectronics	35.2%	39.2%	39.9%	38.7%	37.1%	41.7%	47.3%	47.9%	39.3%	33.9%	35.9%	38.8%	42.2%
Melexis	45.7%	46.0%	45.9%	40.3%	39.0%	42.5%	44.8%	45.6%	43.0%	38.6%	40.8%	42.0%	42.9%
NXP Semiconductor	42.8%	49.9%	51.6%	52.0%	49.2%	54.8%	56.9%	56.9%	56.4%	54.7%	57.5%	58.6%	60.0%
Analog Devices	66.0%	61.1%	68.1%	66.4%	66.4%	57.8%	65.6%	62.5%	57.2%	62.8%	68.1%	68.9%	69.1%
Texas Instruments	61.6%	64.3%	65.1%	63.7%	64.1%	67.5%	68.8%	62.9%	58.1%	57.0%	60.9%	62.7%	63.6%
Microchip	49.9%	60.3%	54.3%	61.6%	61.7%	64.5%	67.1%	67.0%	58.0%	55.4%	64.2%	65.0%	64.3%
Navitas Semiconductor					31.4%	45.0%	31.5%	39.1%	34.0%	31.0%	39.3%	40.4%	41.2%
ON Semiconductor	33.2%	36.7%	38.1%	35.8%	32.7%	40.3%	49.0%	47.1%	45.4%	33.1%	40.1%	43.8%	46.2%
SG Micro	40.2%	43.4%	45.9%	46.9%	48.7%	55.5%	59.0%	49.6%	51.5%	50.8%	51.5%	51.4%	51.4%
Silergy	47.6%	47.7%	46.7%	47.5%	48.1%	53.2%	52.6%	42.7%	53.8%	51.6%	51.8%	53.3%	53.6%
Renesas	43.6%	46.8%	44.2%	42.9%	47.3%	53.2%	57.4%	57.0%	56.1%	57.6%	58.3%	57.9%	58.1%
Innoscence					-266.1%	-289.4%	-61.1%	-19.5%	7.3%	17.7%	33.1%	41.5%	
Yangjie Technology	35.4%	35.6%	31.4%	29.8%	34.3%	35.1%	36.3%	30.3%	33.1%	34.3%	35.4%	35.7%	36.7%
SICC			25.6%	37.7%	35.3%	28.4%	-5.8%	15.8%	25.9%	13.1%	23.2%	23.1%	22.1%
StarPower	28.0%	30.6%	29.4%	30.6%	31.6%	36.7%	40.3%	37.5%	31.6%	26.1%	22.8%	25.0%	27.1%
Allegro Microsystems				44.6%	51.8%	55.6%	56.1%	47.2%	45.0%	48.7%	51.2%	52.4%	
Silan Microelectronics	24.7%	26.7%	25.5%	19.5%	22.5%	33.2%	29.4%	22.2%	19.1%	18.8%	21.5%	24.1%	24.7%
YY bps change													
Infineon		120bps	150bps	-200bps	-380bps	670bps	490bps	30bps	-490bps	-50bps	266bps	358bps	-
STMicroelectronics		397bps	76bps	-127bps	-160bps	466bps	560bps	60bps	-859bps	-546bps	203bps	285bps	345bps
Melexis		30bps	-14bps	-557bps	-129bps	350bps	231bps	83bps	-261bps	-444bps	217bps	120bps	99bps
NXP Semiconductor		706bps	167bps	45bps	-284bps	566bps	209bps	-4bps	-45bps	-170bps	272bps	117bps	141bps
Analog Devices		-483bps	697bps	-166bps	-6bps	-856bps	781bps	-315bps	-531bps	568bps	528bps	76bps	19bps
Texas Instruments		263bps	85bps	-140bps	39bps	337bps	129bps	-586bps	-476bps	-112bps	391bps	181bps	87bps
Microchip		1039bps	-597bps	728bps	96ps	285bps	257bps	-4bps	-908bps	-254bps	874bps	87bps	-72bps
Navitas Semiconductor		0bps	0bps	0bps	3135bps	1367bps	-1353bps	761bps	-508bps	299bps	822bps	118bps	77bps
ON Semiconductor		349bps	139bps	-231bps	-312bps	762bps	870bps	-191bps	-165bps	-1232bps	697bps	372bps	241bps
SG Micro		319bps	251bps	94bps	185bps	677bps	348bps	-938bps	193bps	-72bps	67bps	-5bps	-1bps
Silergy		10bps	-101bps	74bps	62bps	519bps	-70bps	-992bps	1119bps	-222bps	16bps	153bps	28bps
Renesas		316bps	-261bps	-130bps	444bps	585bps	426bps	-45bps	-91bps	155bps	67bps	-41bps	24bps
Innoscence		0bps	0bps	0bps	0bps	-26612bps	-2329bps	22832bps	4151bps	2680bps	1038bps	1537bps	840bps
Yangjie Technology		22bps	-421bps	-156bps	447bps	83bps	119bps	-603bps	282bps	119bps	111bps	29bps	101bps
SICC				1211bps	-240bps	-685bps	-3419bps	2157bps	1008bps	-1284bps	1011bps	-2bps	-107bps
StarPower		263bps	-119bps	120bps	95bps	516bps	358bps	-280bps	-595bps	-547bps	-325bps	217bps	214bps
Allegro Microsystems		0bps	0bps	446bps	713bps	380bps	49bps	-883bps	-226bps	370bps	251bps	122bps	122bps
Silan Microelectronics		204bps	-125bps	-598bps	303bps	1069bps	-374bps	-723bps	-312bps	-26bps	262bps	265bps	56bps
FCF													
Infineon (Enn, reported)	490	603	619	39	(6,727)	1,573	1,647	1,158	23	(1,051)	1,978	3,665	
NXP Semiconductor (\$mn)	1,914	1,895	3,758	1,847	2,090	2,310	2,832	2,686	2,055	2,423	3,715	4,749	5,126
Analog Devices (\$mn)	1,243	948	2,143	1,992	1,909	2,776	4,261	3,242	3,184	4,560	5,753	7,825	8,688
Texas Instruments (\$mn)	4,083	4,668	6,058	5,802	5,490	6,294	5,923	1,349	1,498	2,603	7,923	9,506	12,540
Microchip (\$mn)	982	1,334	1,834	1,958	2,126	2,552	3,254	3,406	1,309	1,062	2,225	2,700	2,939
Navitas Semiconductor (\$mn)				(21)	(44)	(49)	(45)	(66)	(44)	(79)	(40)	(54)	
ON Semiconductor (\$mn)	371	707	759	160	501	1,337	1,628	402	1,212	1,419	1,245	1,578	2,127
SG Micro (RMB mn)	78	124	84	145	324	763	948	(63)	309	466	687	893	1,192
Silergy (NT\$ bn)	1,375	1,765	2,033	2,270	3,651	5,883	6,410	2,873	402	244	3,395	5,272	6,894
Renesas (Yen)	57	(268)	91	(540)	184	(356)	382	229	(944)	328	579	436	
Innoscence (RMB mn)					(562)	(936)	(594)	(336)	(1,017)	1,586	(1,326)	(225)	
Yangjie Technology (RMB mn)	27	(84)	(321)	116	178	(457)	(39)	443	528	685	121	1,140	1,415
SICC (RMB mn)			(35)	(559)	(689)	248	(947)	(1,277)	(445)	479	(949)	(381)	(369)
StarPower (RMB mn)	(25)	(24)	68	39	(215)	(20)	(634)	(1,820)	(1,040)	(94)	212	563	601
Allegro Microsystems (\$mn)				60	105	120	76	5	121	115	195	325	
Silan Microelectronics (RMB mn)	(440)	(807)	(769)	(625)	(495)	12	(1,362)	(803)	(1,239)	(134)	58	729	1,068
Inventory Days													
Infineon	109	104	122	128	115	121	142	178	181	181	162	140	140
STMicroelectronics	97	89	90	100	99	92	97	106	123	137	128	121	127
Melexis	97	97	102	102	86	87	114	136	156	169	155	145	140
NXP Semiconductor	75	97	102	102	86	87	114	136	156	169	155	145	140
Analog Devices	108	93	109	111	115	100	129	131	135	148	134	126	119
Texas Instruments	126	132	145	138	136	115	159	221	249	228	188	171	157
Microchip	100	115	112	128	120	123	160	170	248	198	182	183	175
Navitas Semiconductor					153	335	268	175	103	153	332	283	275
ON Semiconductor	146	110	121	126	127	124	138	174	209	194	198	196	179
StarPower	128	138	111	133	141	134	159	201	201	205	205	205	205
Allegro Microsystems				103	79	107	126	170	139	128	128	125	117
Silan Microelectronics	137	128	163	192	155	125	156	171	153	134	134	134	134
ROIC													
Infineon	17.1%	19.8%	18.6%	13.7%	8.2%	12.8%	15.8%	16.8%	11.6%	9.6%	12.1%	16.6%	19.0%
STMicroelectronics	6.7%	17.7%	23.3%	15.9%	15.4%	24.1%	34.7%	29.9%	9.7%	2.7%	7.1%	14.2%	17.9%
Melexis	57.3%	38.2%	20.7%	24.1%	39.3%	50.9%	44.6%	33.0%	16.9%	21.2%	24.4%	27.6%	
NXP Semiconductor	10.4%	13.1%	14.6%	12.2%	11.0%	0.0%	0.0%	21.4%	17.6%	14.8%	19.0%	22.7%	24.8%
Analog Devices	37.3%	11.3%	11.8%	10.1%	11.1%	6.8%	12.1%	10.7%	7.9%	11.0%	17.1%	19.5%	21.4%
Texas Instruments	32.9%	41.0%	52.5%	47.5%	52.0%	62.0%	57.9%	30.6%	18.8%	19.9%	31.0%	35.4%	41.7%
Microchip	13.7%	20.4%	10.3%	10.1%	11.5%	16.3%	22.6%	23.0%	7.1%	4.9%	13.5%	17.5%	19.9%
Navitas Semiconductor					-436.2%	-332.2%	-35.2%	-33.2%	-33.3%	-28.8%	-35.3%	-38.8%	-50.2%
ON Semiconductor	9.0%	14.3%	17.8%	11.0%	7.3%	20.1%	33.8%	24.0%	17.0%	10.8%	14.4%	19.2%	24.3%
SG Micro	186.2%	23.0%	12.9%	15.5%	31.0%	45.8%	47.1%	13.6%	18.7%	15.5%	24.2%	29.4%	34.0%
Silergy	21.1%	25.4%	21.8%	24.7%	33.3%	52.3%	35.2%	-3.7%	10.8%	6.9%	12.8%	23.4%	28.9%
Innoscence					-23.4%	-28.2%	-27.3%	-29.1%	-15.2%	-17.7%	1.4%	19.4%	
Yangjie Technology	10.3%	12.2%	9.4%	9.5%	15.1%	18.8%	17.4%	9.5%	11.6%	13.9%	17.0%	19.7%	22.4%
SICC			-0.4%	-19.0%	-34.3%	0.0%	-8.8%	-2.2%	2.1%	-3.5%	2.4%	3.7%	4.7%
StarPower	5.9%	11.1%	18.4%	21.0%	13.8%	17.3%	19.2%	14.5%	6.6%	4.3%	4.5%	7.6%	11.0%
Allegro Microsystems				5.7%	24.1%	24.2%	19.6%	4.8%	4.4%	11.2%	18.5%	26.3%	
Silan Microelectronics	1.6%	2.8%	1.5%	-2.1%	0.5%	13.9%	9.1%	4.8%	2.1%	4.2%	8.1%	10.5%	12.5%
ROCE													
Infineon	9.6%	13.0%	17.2%	8.7%	3.5%	9.0%	15.2%	16.0%	7.3%	6.3%	16.2%	22.3%	25.7%
STMicroelectronics	8.5%	22.9%	28.4%	21.5%	19.5%	31.4%	43.6%	36.5%	12.9%	4.1%	8.8%	18.7%	23.8%
Melexis		43.0%	38.4%	19.4%	20.0%	37.9%	46.0%	35.2%	27.8%	16.8%	19.8%	21.5%	23.5%
NXP Semiconductor	-0.7%	9.9%	14.9%	8.5%	2.3%	14.0%	19.0%	18.1%	16.1%	13.4%	21.2%	21.8%	
Analog Devices	11.7%	6.1%	9.2%	7.8%	6.5%	3.3%	6.3%	7.2%	4.3%	7.9%	15.3%	18.0%	20.0%
Texas Instruments	32.4%	38.3%	43.6%	34.9%	33.7%	39.6%	41.0%	24.7%	16.8%	18.9%	27.0%	30.0%	31.4%
Microchip	3.1%	11.5%	3.9%	5.1%	6.1%	10.8%	19.4%	24.6%	4.6%	1.3%	8.8%	13.3%	14.0%
Navitas Semiconductor					-44.6%	-24.1%	-29.9%	-27.2%	-35.4%	-24.0%	-14.8%	-15.5%	-16.2%
ON Semiconductor	4.4%	11.8%	13.9%	6.6%	5.0%	15.9%	23.8%	23.0%	13.9%	0.8%	7.7%	15.4%	21.5%
SG Micro	35.7%	21.0%	11.8%	15.3%	20.5%	34.6%	34.5%	7.4%	12.1%	10.6%	15.2%	18.3%	20.2%
Silergy	19.7%	17.5%	15.8%	16.5%	18.1%	31.0%	20.7%	-1.6%	6.5%	5.5%	8.5%	14.7%	17.2%
Innoscence					-87.1%	-23.2%	-23.2%	-15.3%	-9.6				

Prices for stocks mentioned (in local currency, last close): Infineon Technologies (IFX-DE) 55.97, STMicroelectronics (STMMI.IT) 43.10, Analog Devices (ADI-US) 373.09, NXP Semiconductors (NXPI-US) 225.56, Allegro MicroSystems (ALGM-US) 37.16, Renesas Electronics (6723-JP) 3,433.00, Yangjie Technology (300373-CN) 103.04, SICC (688234-CN) 109.95, Melexis (MELE-BE) 65.00, Microchip Technology (MCHP-US) 76.08, Innoscience (2577-HK) 46.74, SG Micro (300661-CN) 116.60, StarPower Semiconductor (603290-CN) 97.96, Texas Instruments (TXN-US) 264.36, Navitas Semiconductor (NVTX-US) 12.97, Silergy (6415-TW) 445.50, Silan Microelectronics (600460-CN) 35.89.

Morgan Stanley is acting as financial advisor to onsemi in connection with its definitive agreement to acquire Synaptics Incorporated ("Synaptics"), as announced on June 25, 2026. The proposed transaction is subject to approval by Synaptics' shareholders, receipt of required regulatory approval and other customary closing conditions. This report and the information provided herein is not intended to (i) provide voting advice, (ii) serve as an endorsement of the proposed transaction, or (iii) result in the procurement, withholding or revocation of a proxy or any other action by a security holder. Onsemi has agreed to pay fees to Morgan Stanley for its financial services. Please refer to the notes at the end of the report.

Disclosure Section

The information and opinions in Morgan Stanley Research were prepared or are disseminated by Morgan Stanley & Co. LLC and/or Morgan Stanley C.T.V.M. S.A. and/or Morgan Stanley México, Casa de Bolsa, S.A. de C.V. and/or Morgan Stanley Canada Limited and/or Morgan Stanley & Co. International plc and/or Morgan Stanley Europe S.E. and/or RMB Morgan Stanley Proprietary Limited and/or Morgan Stanley MUFJ Securities Co., Ltd. and/or Morgan Stanley Capital Group Japan Co., Ltd. and/or Morgan Stanley Asia Limited and/or Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research) and/or Morgan Stanley Taiwan Limited and/or Morgan Stanley & Co International plc, Seoul Branch, and/or Morgan Stanley Australia Limited (A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents), and/or Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents), and/or Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105), Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com which accepts the responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research, and their affiliates (collectively, "Morgan Stanley"). Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts.

For important disclosures, stock price charts and equity rating histories regarding companies that are the subject of this report, please see the Morgan Stanley Research Disclosure Website at www.morganstanley.com/eqr/disclosures/webapp/generalresearch, or contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY, 10036 USA.

For valuation methodology and risks associated with any recommendation, rating or price target referenced in this research report, please contact the Client Support Team as follows: US/Canada +1 800 303-2495; Hong Kong +852 2848-5999; Latin America +1 718 754-5444 (U.S.); London +44 (0)20-7425-8169; Singapore +65 6834-6860; Sydney +61 (0)2-9770-1505; Tokyo +81 (0)3-6836-9000. Alternatively you may contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY 10036 USA.

Analyst Certification

The following analysts hereby certify that their views about the companies and their securities discussed in this report are accurately expressed and that they have not received and will not receive direct or indirect compensation in exchange for expressing specific recommendations or views in this report: Charlie Chan; Daisy Dai, CFA; Shawn Kim; Nicole Kozhukhov; Joseph Moore; Nigel van Putten; Amelia M Scicluna; Lee Simpson; Hidetaka Suzuki; Lucas Wang; Kazuo Yoshikawa, CFA; Henry Zhao.

Global Research Conflict Management Policy

Morgan Stanley Research has been published in accordance with our conflict management policy, which is available at www.morganstanley.com/institutional/research/conflict/policies. A Portuguese version of the policy can be found at www.morganstanley.com.br

Important Regulatory Disclosures on Subject Companies

A family member of an author of this report has an investor relations role at Qualcomm Inc.

As of July 31, 2026, Morgan Stanley beneficially owned 1% or more of a class of common equity securities of the following companies covered in Morgan Stanley Research: ACM Research Inc, Advanced Micro Devices, Advanced Micro-Fabrication Equipment Inc, Advanced Wireless Semiconductor Co, Aeva Technologies Inc, Aixtron SE, Alchip Technologies Ltd, AllRing Tech Co., Ambarella Inc, Analog Devices Inc., AP Memory Technology Corp, Arm Holdings plc, ASE Technology Holding Co. Ltd., ASML Holding NV, ASMPT Ltd, Aspeed Technology, Astera Labs Inc, Broadcom Inc., Cadence Design Systems Inc, Cambricon Technology Corporation, Cerebras Systems, eMemory Technology Inc, GigaDevice Semiconductor Beijing Inc, GlobalWafers Co Ltd, Gudeng Precision, Hangzhou Silan Microelectronics Co. Ltd., HOYA, Hua Hong Semiconductor Ltd, Infineon Technologies AG, Intel Corporation, IonQ Inc, King Yuan Electronics Co Ltd, KIOXIA Holdings, Macronix International Co Ltd, Marvell Technology Group Ltd, MediaTek, Microchip Technology Inc., Micron Technology Inc., Montage Technology Co Ltd, NAURA Technology Group Co Ltd, Navitas Semiconductor Corp, Nordic Semiconductor ASA, Nuvoton Technology Corporation, NVIDIA Corp., NXP Semiconductor NV, ON Semiconductor Corp., Phison Electronics Corp, Powerchip Semiconductor Manufacturing Co, Qorvo Inc, Qualcomm Inc., Renesas Electronics, Rohm, SanDisk Corporation., Semtech Corp., SG Micro Corp., Shanghai Fudan Microelectronics, Silergy Corp., Silicon Laboratories Inc., Silicon Motion, Skyworks Solutions Inc, Socionext, Soitec SA, StarPower Semiconductor Ltd, STMicroelectronics NV, Synopsys Inc., Texas Instruments, TSMC, UMC, Vanguard International Semiconductor, WIN Semiconductors Corp, Winbond Electronics Corp, WinWay Technology Co Ltd, Wolfspeed, INC, WPG Holdings, WT Microelectronics Co. Ltd..

Within the last 12 months, Morgan Stanley managed or co-managed a public offering (or 144A offering) of securities of Advanced Micro Devices, Aeva Technologies Inc, Aixtron SE, Alchip Technologies Ltd, Broadcom Inc., Cerebras Systems, GlobalFoundries Inc, Iluvatar CoreX Semiconductor Co., Ltd., Intel Corporation, Montage Technology Co Ltd, NVIDIA Corp., ON Semiconductor Corp., Powerchip Semiconductor Manufacturing Co, Quantinuum, Semtech Corp., STMicroelectronics NV.

Within the last 12 months, Morgan Stanley has received compensation for investment banking services from Advanced Micro Devices, Aeva Technologies Inc, Aixtron SE, Allegro Microsystems Inc, Amkor Technology Inc, Analog Devices Inc., ASM International NV, ASMPT Ltd, BE Semiconductor Industries NV, Broadcom Inc., Cerebras Systems, HOYA, Iluvatar CoreX Semiconductor Co., Ltd., Intel Corporation, IonQ Inc, Micron Technology Inc., Montage Technology Co Ltd, Navitas Semiconductor Corp, NVIDIA Corp., NXP Semiconductor NV, ON Semiconductor Corp., Powerchip Semiconductor Manufacturing Co, Quantinuum, Semtech Corp., Silicon Motion, STMicroelectronics NV, Texas Instruments.

In the next 3 months, Morgan Stanley expects to receive or intends to seek compensation for investment banking services from Advanced Micro Devices, Advanced Micro-Fabrication Equipment Inc, Aeva Technologies Inc, Aixtron SE, Alchip Technologies Ltd, Allegro Microsystems Inc, Ambarella Inc, Amkor Technology Inc, Analog Devices Inc., AP Memory Technology Corp, Arm Holdings plc, ASE Technology Holding Co. Ltd., ASM International NV, ASMedia Technology Inc, ASML Holding NV, ASMPT Ltd, Aspeed Technology, Astera Labs Inc, BE Semiconductor Industries NV, Broadcom Inc., Cadence Design Systems Inc, Cerebras Systems, Espressif Systems, GigaDevice Semiconductor Beijing Inc, GlobalFoundries Inc, GlobalWafers Co Ltd, Gudeng Precision, Himax Technologies Inc, HOYA, Hua Hong Semiconductor Ltd, Iluvatar CoreX Semiconductor Co., Ltd., Infineon Technologies AG, Innoscience, Intel Corporation, IonQ Inc, King Yuan Electronics Co Ltd, KIOXIA Holdings, Macronix International Co Ltd, Marvell Technology Group Ltd, MediaTek, Melexis N.V., MetaX Integrated Circuits, Microchip Technology Inc., Micron Technology Inc., Montage Technology Co Ltd, Navitas Semiconductor Corp, Nordic Semiconductor ASA, Novatek, NVIDIA Corp., NXP Semiconductor NV, ON Semiconductor Corp., Phison Electronics Corp, Powerchip Semiconductor Manufacturing Co, Qualcomm Inc., Quantinuum, Realtek Semiconductor, Renesas Electronics, Rohm, SanDisk Corporation., Semtech Corp., SG Micro Corp., Shenzhen Longsys Electronics Co Ltd, SICC Co Ltd, Silergy Corp., Silicon Laboratories Inc., Silicon Motion, Skyworks Solutions Inc, Socionext, Soitec SA, STMicroelectronics NV, Synopsys Inc., Texas

Instruments, TSMC, UMC, Universal Scientific Ind. (Shanghai), Vanguard International Semiconductor, VAT Group AG, Winbond Electronics Corp, WinWay Technology Co Ltd, Wolfspeed, INC, WPG Holdings, WT Microelectronics Co. Ltd..

Within the last 12 months, Morgan Stanley has received compensation for products and services other than investment banking services from Advanced Micro Devices, Allegro Microsystems Inc, Ambarella Inc, Amkor Technology Inc, Analog Devices Inc., ASE Technology Holding Co. Ltd., ASM International NV, BE Semiconductor Industries NV, Broadcom Inc., Cadence Design Systems Inc, Cerebras Systems, GlobalFoundries Inc, Infineon Technologies AG, Intel Corporation, King Yuan Electronics Co Ltd, Marvell Technology Group Ltd, MediaTek, Microchip Technology Inc., Micron Technology Inc., Nanya Technology Corp., Novatek, Nuvoton Technology Corporation, NVIDIA Corp., NXP Semiconductor NV, ON Semiconductor Corp., Qorvo Inc, Qualcomm Inc., Realtek Semiconductor, Semtech Corp., Silicon Laboratories Inc., Silicon Motion, SMIC, Synopsys Inc., Texas Instruments, TSMC, UMC, Universal Scientific Ind. (Shanghai), Winbond Electronics Corp, WT Microelectronics Co. Ltd..

Within the last 12 months, Morgan Stanley has provided or is providing investment banking services to, or has an investment banking client relationship with, the following company: Advanced Micro Devices, Advanced Micro-Fabrication Equipment Inc, Aeva Technologies Inc, Aixtron SE, Alchip Technologies Ltd, Allegro Microsystems Inc, Ambarella Inc, Amkor Technology Inc, Analog Devices Inc., AP Memory Technology Corp, Arm Holdings plc, ASE Technology Holding Co. Ltd., ASM International NV, ASMedia Technology Inc, ASML Holding NV, ASMPT Ltd, Aspeed Technology, Astera Labs Inc, BE Semiconductor Industries NV, Broadcom Inc., Cadence Design Systems Inc, Cerebras Systems, Espressif Systems, GigaDevice Semiconductor Beijing Inc, GlobalFoundries Inc, GlobalWafers Co Ltd, Gudeng Precision, Himax Technologies Inc, HOYA, Hua Hong Semiconductor Ltd, Iluvatar CoreX Semiconductor Co., Ltd., Infineon Technologies AG, Innoscience, Intel Corporation, IonQ Inc, King Yuan Electronics Co Ltd, KIOXIA Holdings, Macronix International Co Ltd, Marvell Technology Group Ltd, MediaTek, Melexis N.V., MetaX Integrated Circuits, Microchip Technology Inc., Micron Technology Inc., Montage Technology Co Ltd, Navitas Semiconductor Corp, Nordic Semiconductor ASA, Novatek, NVIDIA Corp., NXP Semiconductor NV, ON Semiconductor Corp., Phison Electronics Corp, Powerchip Semiconductor Manufacturing Co, Qualcomm Inc., Quantinuum, Realtek Semiconductor, Renesas Electronics, Rohm, SanDisk Corporation., Semtech Corp., SG Micro Corp., Shenzhen Longsys Electronics Co Ltd, SICC Co Ltd, Silergy Corp., Silicon Laboratories Inc., Silicon Motion, Skyworks Solutions Inc, Socionext, Soitec SA, STMicroelectronics NV, Synopsys Inc., Texas Instruments, TSMC, UMC, Universal Scientific Ind. (Shanghai), Vanguard International Semiconductor, VAT Group AG, Winbond Electronics Corp, WinWay Technology Co Ltd, Wolfspeed, INC, WPG Holdings, WT Microelectronics Co. Ltd..

Within the last 12 months, Morgan Stanley has either provided or is providing non-investment banking, securities-related services to and/or in the past has entered into an agreement to provide services or has a client relationship with the following company: Advanced Micro Devices, Aixtron SE, Allegro Microsystems Inc, Ambarella Inc, Amkor Technology Inc, Analog Devices Inc., ASE Technology Holding Co. Ltd., ASM International NV, BE Semiconductor Industries NV, Broadcom Inc., Cadence Design Systems Inc, Cerebras Systems, GlobalFoundries Inc, Iluvatar CoreX Semiconductor Co., Ltd., Infineon Technologies AG, Intel Corporation, IonQ Inc, King Yuan Electronics Co Ltd, Marvell Technology Group Ltd, MediaTek, Microchip Technology Inc., Micron Technology Inc., Montage Technology Co Ltd, Nanya Technology Corp., Novatek, Nuvoton Technology Corporation, NVIDIA Corp., NXP Semiconductor NV, ON Semiconductor Corp., Qorvo Inc, Qualcomm Inc., Realtek Semiconductor, Renesas Electronics, SanDisk Corporation., Semtech Corp., Silicon Laboratories Inc., Silicon Motion, SMIC, Synopsys Inc., Texas Instruments, TSMC, UMC, Universal Scientific Ind. (Shanghai), Winbond Electronics Corp, Wolfspeed, INC, WT Microelectronics Co. Ltd..

Morgan Stanley & Co. LLC makes a market in the securities of Aeva Technologies Inc, Ambarella Inc.

The equity research analysts or strategists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality of research, investor client feedback, stock picking, competitive factors, firm revenues and overall investment banking revenues. Equity Research analysts' or strategists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

Morgan Stanley and its affiliates do business that relates to companies/instruments covered in Morgan Stanley Research, including market making, providing liquidity, fund management, commercial banking, extension of credit, investment services and investment banking. Morgan Stanley sells to and buys from customers the securities/instruments of companies covered in Morgan Stanley Research on a principal basis. Morgan Stanley may have a position in the debt of the Company or instruments discussed in this report. Morgan Stanley trades or may trade as principal in the debt securities (or in related derivatives) that are the subject of the debt research report.

Certain disclosures listed above are also for compliance with applicable regulations in non-US jurisdictions.

STOCK RATINGS

Morgan Stanley uses a relative rating system using terms such as Overweight, Equal-weight, Not-Rated or Underweight (see definitions below). Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold and sell. Investors should carefully read the definitions of all ratings used in Morgan Stanley Research. In addition, since Morgan Stanley Research contains more complete information concerning the analyst's views, investors should carefully read Morgan Stanley Research, in its entirety, and not infer the contents from the rating alone. In any case, ratings (or research) should not be used or relied upon as investment advice. An investor's decision to buy or sell a stock should depend on individual circumstances (such as the investor's existing holdings) and other considerations.

Global Stock Ratings Distribution

(as of July 31, 2026)

The Stock Ratings described below apply to Morgan Stanley's Fundamental Equity Research and do not apply to Debt Research produced by the Firm.

For disclosure purposes only (in accordance with FINRA requirements), we include the category headings of Buy, Hold, and Sell alongside our ratings of Overweight, Equal-weight, Not-Rated and Underweight. Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold, and sell but represent recommended relative weightings (see definitions below). To satisfy regulatory requirements, we correspond Overweight, our most positive stock rating, with a buy recommendation; we correspond Equal-weight and Not-Rated to hold and Underweight to sell recommendations, respectively.

Coverage Universe			Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
Stock Rating Category	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1529	42%	442	47%	29%	750	43%
Equal-weight/Hold	1582	43%	396	42%	25%	773	44%
Not-Rated/Hold	3	0%	1	0%	33%	1	0%
Underweight/Sell	560	15%	97	10%	17%	217	12%
Total	3,674		936			1741	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

Analyst Stock Ratings

Overweight (O or Over) - The stock's total return is expected to exceed the total return of the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis over the next 12-18 months.

Equal-weight (E or Equal) - The stock's total return is expected to be in line with the total return of the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis over the next 12-18 months.

Not-Rated (NR) - Currently the analyst does not have adequate conviction about the stock's total return relative to the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Underweight (U or Under) - The stock's total return is expected to be below the total return of the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.

In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.

Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.

Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Important Disclosures for Morgan Stanley Smith Barney LLC Customers

Important disclosures regarding any material conflict of interest that can reasonably be expected to have influenced Morgan Stanley Smith Barney LLC's choice of a third-party research provider or the subject company of a third-party research report, are available on the Morgan Stanley Wealth Management disclosure website at www.morganstanley.com/online/researchdisclosures. For Morgan Stanley specific disclosures, you may refer to <https://www.morganstanley.com/eqr/disclosures/webapp/generalresearch>.

Each Morgan Stanley research report is reviewed and approved on behalf of Morgan Stanley Smith Barney LLC. This review and approval is conducted by the same person who reviews the research report on behalf of Morgan Stanley. This could create a conflict of interest.

Other Important Disclosures

A member of Research who had or could have had access to the research prior to completion owns securities (or related derivatives) in the Analog Devices Inc., Broadcom Inc., Microchip Technology Inc., NVIDIA Corp., Qualcomm Inc. This person is not a research analyst or a member of research analyst's household.

Morgan Stanley Research policy is to update research reports as and when the Research Analyst and Research Management deem appropriate, based on developments with the issuer, the sector, or the market that may have a material impact on the research views or opinions stated therein. In addition, certain Research publications are intended to be updated on a regular periodic basis (weekly/monthly/quarterly/annual) and will ordinarily be updated with that frequency, unless the Research Analyst and Research Management determine that a different publication schedule is appropriate based on current conditions.

Morgan Stanley is not acting as a municipal advisor and the opinions or views contained herein are not intended to be, and do not constitute, advice within the meaning of Section 975 of the Dodd-Frank Wall Street Reform and Consumer Protection Act.

Morgan Stanley produces an equity research product called a "Tactical Idea." Views contained in a "Tactical Idea" on a particular stock may be contrary to the recommendations or views expressed in research on the same stock. This may be the result of differing time horizons, methodologies, market events, or other factors. For all research available on a particular stock, please contact your sales representative or go to Matrix at <http://www.morganstanley.com/matrix>.

Morgan Stanley Research is provided to our clients through our proprietary research portal on Matrix and also distributed electronically by Morgan Stanley to clients. Certain, but not all, Morgan Stanley Research products are also made available to clients through third-party vendors or redistributed to clients through alternate electronic means as a convenience. For access to all available Morgan Stanley Research, please contact your sales representative or go to Matrix at <http://www.morganstanley.com/matrix>.

Any access and/or use of Morgan Stanley Research is subject to Morgan Stanley's Terms of Use (<http://www.morganstanley.com/terms.html>). By accessing and/or using Morgan Stanley Research, you are indicating that you have read and agree to be bound by our Terms of Use (<http://www.morganstanley.com/terms.html>). In addition you consent to Morgan Stanley processing your personal data and using cookies in accordance with our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html), including for the purposes of setting your preferences and to collect readership data so that we can deliver better and more personalized service and products to you. To find out more information about how Morgan Stanley processes personal data, how we use cookies and how to reject cookies see our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html). Please use the provided link to review the Terms and Conditions and Most Important Terms and Conditions for Morgan Stanley India Company Private Limited (https://www.morganstanley.com/assets/pdfs/about-us-global-offices/india/Terms_and_conditions.pdf) and the following link to review the audit report (<https://ny.matrix.ms.com/eqr/research/webapp/researchdocs/>

[MSICPL_Morgan_Stanley_Research_Audit_Report.pdf](#)).

If you do not agree to our Terms of Use and/or if you do not wish to provide your consent to Morgan Stanley processing your personal data or using cookies please do not access our research. Morgan Stanley Research does not provide individually tailored investment advice. Morgan Stanley Research has been prepared without regard to the circumstances and objectives of those who receive it. Morgan Stanley recommends that investors independently evaluate particular investments and strategies, and encourages investors to seek the advice of a financial adviser. The appropriateness of an investment or strategy will depend on an investor's circumstances and objectives. The securities, instruments, or strategies discussed in Morgan Stanley Research may not be suitable for all investors, and certain investors may not be eligible to purchase or participate in some or all of them. Morgan Stanley Research is not an offer to buy or sell or the solicitation of an offer to buy or sell any security/instrument or to participate in any particular trading strategy. The value of and income from your investments may vary because of changes in interest rates, foreign exchange rates, default rates, prepayment rates, securities/instruments prices, market indexes, operational or financial conditions of companies or other factors. There may be time limitations on the exercise of options or other rights in securities/instruments transactions. Past performance is not necessarily a guide to future performance. Estimates of future performance are based on assumptions that may not be realized. If provided, and unless otherwise stated, the closing price on the cover page is that of the primary exchange for the subject company's securities/instruments.

The fixed income research analysts, strategists or economists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality, accuracy and value of research, firm profitability or revenues (which include fixed income trading and capital markets profitability or revenues), client feedback and competitive factors. Fixed Income Research analysts', strategists' or economists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

The "Important Regulatory Disclosures on Subject Companies" section in Morgan Stanley Research lists all companies mentioned where Morgan Stanley owns 1% or more of a class of common equity securities of the companies. For all other companies mentioned in Morgan Stanley Research, Morgan Stanley may have an investment of less than 1% in securities/instruments or derivatives of securities/instruments of companies and may trade them in ways different from those discussed in Morgan Stanley Research. Employees of Morgan Stanley not involved in the preparation of Morgan Stanley Research may have investments in securities/instruments or derivatives of securities/instruments of companies mentioned and may trade them in ways different from those discussed in Morgan Stanley Research. Derivatives may be issued by Morgan Stanley or associated persons.

With the exception of information regarding Morgan Stanley, Morgan Stanley Research is based on public information. Morgan Stanley makes every effort to use reliable, comprehensive information, but we make no representation that it is accurate or complete. We have no obligation to tell you when opinions or information in Morgan Stanley Research change apart from when we intend to discontinue equity research coverage of a subject company. Facts and views presented in Morgan Stanley Research have not been reviewed by, and may not reflect information known to, professionals in other Morgan Stanley business areas, including investment banking personnel.

Morgan Stanley Research personnel may participate in company events such as site visits and are generally prohibited from accepting payment by the company of associated expenses unless pre-approved by authorized members of Research management.

Morgan Stanley may make investment decisions that are inconsistent with the recommendations or views in this report.

To our readers based in Taiwan or trading in Taiwan securities/instruments: Information on securities/instruments that trade in Taiwan is distributed by Morgan Stanley Taiwan Limited ("MSTL"). Such information is for your reference only. The reader should independently evaluate the investment risks and is solely responsible for their investment decisions. Morgan Stanley Research may not be distributed to the public media or quoted or used by the public media without the express written consent of Morgan Stanley. Any non-customer reader within the scope of Article 7-1 of the Taiwan Stock Exchange Recommendation Regulations accessing and/or receiving Morgan Stanley Research is not permitted to provide Morgan Stanley Research to any third party (including but not limited to related parties, affiliated companies and any other third parties) or engage in any activities regarding Morgan Stanley Research which may create or give the appearance of creating a conflict of interest. Information on securities/instruments that do not trade in Taiwan is for informational purposes only and is not to be construed as a recommendation or a solicitation to trade in such securities/instruments. MSTL may not execute transactions for clients in these securities/instruments.

Certain information in Morgan Stanley Research was sourced by employees of the Shanghai Representative Office of Morgan Stanley Asia Limited for the use of Morgan Stanley Asia Limited. Morgan Stanley is not incorporated under PRC law and the research in relation to this report is conducted outside the PRC. Morgan Stanley Research does not constitute an offer to sell or the solicitation of an offer to buy any securities in the PRC. PRC investors shall have the relevant qualifications to invest in such securities and shall be responsible for obtaining all relevant approvals, licenses, verifications and/or registrations from the relevant governmental authorities themselves. Neither this report nor any part of it is intended as, or shall constitute, provision of any consultancy or advisory service of securities investment as defined under PRC law. Such information is provided for your reference only.

Morgan Stanley Research is disseminated in Brazil by Morgan Stanley C.T.V.M. S.A. located at Av. Brigadeiro Faria Lima, 3600, 6th floor, São Paulo - SP, Brazil; and is regulated by the Comissão de Valores Mobiliários; in Mexico by Morgan Stanley México, Casa de Bolsa, S.A. de C.V. which is regulated by Comisión Nacional Bancaria y de Valores. Paseo de los Tamarindos 90, Torre 1, Col. Bosques de las Lomas Floor 29, 05120 Mexico City; in Japan by Morgan Stanley MUFG Securities Co., Ltd. and, for Commodities related research reports only, Morgan Stanley Capital Group Japan Co., Ltd; in Hong Kong by Morgan Stanley Asia Limited (which accepts responsibility for its contents) and by Morgan Stanley Bank Asia Limited; in Singapore by Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research) and by Morgan Stanley Bank Asia Limited, Singapore Branch (Registration number T14FC0118); in Australia to "wholesale clients" within the meaning of the Australian Corporations Act by Morgan Stanley Australia Limited A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents; in Australia to "wholesale clients" and "retail clients" within the meaning of the Australian Corporations Act by Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents; in Korea by Morgan Stanley & Co International plc, Seoul Branch; in India by Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105); Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com. Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts; in Canada by Morgan Stanley Canada Limited; in Germany and the European Economic Area where required by Morgan Stanley Europe S.E., authorised and regulated by Bundesanstalt fuer Finanzdienstleistungsaufsicht (BaFin) under the reference number 149169; in the US by Morgan Stanley & Co. LLC, which accepts responsibility for its contents. Morgan Stanley & Co. International plc, authorized by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority, disseminates in the UK research that it has prepared, and research which has been prepared by any of its affiliates, only to persons who (i) are investment professionals falling within Article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (as amended, the "Order"); (ii) are persons who are high net worth entities falling within Article 49(2)(a) to (d) of the Order; or (iii) are persons to whom an invitation or inducement to engage in investment activity (within the meaning of section 21 of the Financial Services and Markets Act 2000, as amended) may otherwise lawfully be communicated or caused to be communicated. RMB Morgan Stanley Proprietary Limited is a member

of the JSE Limited and A2X (Pty) Ltd. RMB Morgan Stanley Proprietary Limited is a joint venture owned equally by Morgan Stanley International Holdings Inc. and RMB Investment Advisory (Proprietary) Limited, which is wholly owned by FirstRand Limited. The information in Morgan Stanley Research is being disseminated by Morgan Stanley Saudi Arabia, regulated by the Capital Market Authority in the Kingdom of Saudi Arabia, and is directed at Sophisticated investors only.

Morgan Stanley Hong Kong Securities Limited is the liquidity provider/market maker for securities of ASM International NV, ASMPT Ltd, GigaDevice Semiconductor Beijing Inc, Hua Hong Semiconductor Ltd, Montage Technology Co Ltd listed on the Stock Exchange of Hong Kong Limited. An updated list can be found on HKEx website: <http://www.hkex.com.hk>.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (DIFC Branch), regulated by the Dubai Financial Services Authority (the DFSA) or by Morgan Stanley & Co. International plc (ADGM Branch), regulated by the Financial Services Regulatory Authority Abu Dhabi (the FSRA), and is directed at Professional Clients only, as defined by the DFSA or the FSRA, respectively. The financial products or financial services to which this research relates will only be made available to a customer who we are satisfied meets the regulatory criteria of a Professional Client. A distribution of the different MS Research ratings or recommendations, in percentage terms for Investments in each sector covered, is available upon request from your sales representative.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (QFC Branch), regulated by the Qatar Financial Centre Regulatory Authority (the QFCRA), and is directed at business customers and market counterparties only and is not intended for Retail Customers as defined by the QFCRA.

As required by the Capital Markets Board of Turkey, investment information, comments and recommendations stated here, are not within the scope of investment advisory activity. Investment advisory service is provided exclusively to persons based on their risk and income preferences by the authorized firms. Comments and recommendations stated here are general in nature. These opinions may not fit to your financial status, risk and return preferences. For this reason, to make an investment decision by relying solely to this information stated here may not bring about outcomes that fit your expectations.

The trademarks and service marks contained in Morgan Stanley Research are the property of their respective owners. Third-party data providers make no warranties or representations relating to the accuracy, completeness, or timeliness of the data they provide and shall not have liability for any damages relating to such data. The Global Industry Classification Standard (GICS) was developed by and is the exclusive property of MSCI and S&P.

Morgan Stanley Research, or any portion thereof may not be reprinted, sold or redistributed without the written consent of Morgan Stanley.

Indicators and trackers referenced in Morgan Stanley Research may not be used as, or treated as, a benchmark under Regulation EU 2016/1011, or any other similar framework.

The issuers and/or fixed income products recommended or discussed in certain fixed income research reports may not be continuously followed. Accordingly, investors should regard those fixed income research reports as providing stand-alone analysis and should not expect continuing analysis or additional reports relating to such issuers and/or individual fixed income products.

Morgan Stanley may hold, from time to time, material financial and commercial interests regarding the company subject to the Research report.

Registration granted by SEBI and certification from the National Institute of Securities Markets (NISM) in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Investment in securities market are subject to market risks. Read all the related documents carefully before investing.

INDUSTRY COVERAGE: Technology - European Semiconductors

COMPANY (TICKER)	RATING (AS OF)	PRICE* (08/24/2026)
Lee Simpson		
ASML Holding NV (ASML.AS)	O (09/22/2025)	€1,489.80
Infineon Technologies AG (IFXGn.DE)	O (02/06/2025)	€54.16
STMicroelectronics NV (STMPA.PA)	O (03/26/2026)	€42.22
Nigel van Putten		
Aixtron SE (AIXGn.DE)	E (05/25/2023)	€36.75
ASM International NV (ASMI.AS)	O (06/19/2024)	€822.40
BE Semiconductor Industries NV (BESI.AS)	O (11/07/2022)	€208.00
Melexis N.V. (MLXS.BR)	E (02/05/2026)	€63.80
Nordic Semiconductor ASA (NOD.OL)	E (02/10/2025)	NKr 159.50
Soitec SA (SOIT.PA)	O (03/26/2026)	€107.85
VAT Group AG (VACN.S)	E (03/21/2025)	SFr 603.40

Stock Ratings are subject to change. Please see latest research for each company.

* Historical prices are not split adjusted.

INDUSTRY COVERAGE: Greater China Technology Semiconductors

COMPANY (TICKER)	RATING (AS OF)	PRICE* (08/24/2026)
Charlie Chan		
ACM Research Inc (ACMR.O)	O (03/07/2023)	US\$77.35
Advanced Micro-Fabrication Equipment Inc (688012.SS)	O (11/06/2023)	Rmb357.65
Advanced Wireless Semiconductor Co (8086.TWO)	U (07/14/2025)	NT\$112.00
Alchip Technologies Ltd (3661.TW)	O (05/14/2021)	NT\$3,735.00
ASE Technology Holding Co. Ltd. (3711.TW)	O (09/15/2024)	NT\$592.00
Cambricon Technology Corporation (688256.SS)	O (04/27/2026)	Rmb969.05

Global Unichip Corp (3443.TW)	O (06/24/2026)	NT\$5,365.00
GlobalWafers Co Ltd (6488.TWO)	E (05/19/2026)	NT\$1,010.00
Gudeng Precision (3680.TWO)	O (11/25/2025)	NT\$464.00
Hua Hong Semiconductor Ltd (1347.HK)	E (03/12/2026)	HK\$108.80
Iluvatar CoreX Semiconductor Co., Ltd. (9903.HK)	O (04/27/2026)	HK\$361.00
King Yuan Electronics Co Ltd (2449.TW)	O (03/03/2023)	NT\$239.00
Maxscend Microelectronics Co Ltd (300782.SZ)	U (01/11/2021)	Rmb74.55
MediaTek (2454.TW)	O (11/28/2025)	NT\$3,765.00
MetaX Integrated Circuits (688802.SS)	E (04/27/2026)	Rmb664.00
Nanya Technology Corp. (2408.TW)	O (05/28/2026)	NT\$501.00
NAURA Technology Group Co Ltd (002371.SZ)	O (11/06/2023)	Rmb707.90
OmniVision Integrated Circuits Group Inc (603501.SS)	E (11/17/2025)	Rmb82.50
Phison Electronics Corp (8299.TWO)	E (02/25/2026)	NT\$2,135.00
SG Micro Corp. (300661.SZ)	E (11/03/2025)	Rmb111.13
Silergy Corp. (6415.TW)	U (05/19/2026)	NT\$450.00
SMIC (0981.HK)	O (10/21/2025)	HK\$66.75
TSMC (2330.TW)	O (02/07/2022)	NT\$2,375.00
UMC (2303.TW)	O (05/19/2026)	NT\$123.50
Vanguard International Semiconductor (5347.TWO)	E (01/14/2026)	NT\$151.00
WIN Semiconductors Corp (3105.TWO)	U (07/14/2025)	NT\$355.00

Daisy Dai, CFA

ASMP Ltd (0522.HK)	O (07/24/2025)	HK\$157.80
China Resources Microelectronics Limited (688396.SS)	U (03/02/2026)	Rmb61.22
Empyrean Technology Co Ltd (301269.SZ)	E (01/17/2025)	Rmb91.25
Hangzhou Silan Microelectronics Co. Ltd. (600460.SS)	U (08/25/2025)	Rmb35.56
Hygon Information Technology Co., Ltd. (688041.SS)	O (07/03/2026)	Rmb230.81
Innoscence (2577.HK)	E (10/13/2025)	HK\$43.48
JCET Group Co Ltd (600584.SS)	E (01/16/2026)	Rmb73.80
Shanghai Fudan Microelectronics (1385.HK)	O (03/07/2025)	HK\$25.34
SICC Co Ltd (688234.SS)	O (03/20/2026)	Rmb105.05
StarPower Semiconductor Ltd (603290.SS)	E (05/14/2026)	Rmb92.99
Unigroup Guoxin Microelectronics Co Ltd (002049.SZ)	U (01/10/2023)	Rmb60.99
Universal Scientific Ind. (Shanghai) (601231.SS)	O (11/05/2025)	Rmb25.90
Yangjie Technology (300373.SZ)	O (06/10/2022)	Rmb97.80

Daniel Yen, CFA

AP Memory Technology Corp (6531.TW)	O (07/11/2025)	NT\$845.00
ASMedia Technology Inc (5269.TW)	U (10/03/2025)	NT\$1,400.00
Aspeed Technology (5274.TWO)	O (06/09/2025)	NT\$15,115.00
Egis Technology Inc (6462.TWO)	E (01/28/2026)	NT\$98.90
Elan Microelectronics Corp (2458.TW)	O (10/03/2025)	NT\$139.00
eMemory Technology Inc (3529.TWO)	O (08/18/2026)	NT\$2,200.00
Espressif Systems (688018.SS)	O (05/15/2023)	Rmb108.31
GigaDevice Semiconductor Beijing Inc (603986.SS)	O (05/15/2025)	Rmb381.66
Macronix International Co Ltd (2337.TW)	O (09/18/2025)	NT\$124.00
Montage Technology Co Ltd (6809.HK)	O (03/18/2026)	HK\$250.00
Montage Technology Co Ltd (688008.SS)	O (03/18/2026)	Rmb191.83
Novatek (3034.TW)	U (02/04/2026)	NT\$542.00
Nuvoton Technology Corporation (4919.TW)	U (11/10/2025)	NT\$116.00
Parade Technologies Ltd (4966.TWO)	O (05/27/2026)	NT\$557.00
Powerchip Semiconductor Manufacturing Co (6770.TW)	O (10/27/2025)	NT\$70.80
Realtek Semiconductor (2379.TW)	E (01/30/2026)	NT\$716.00
Shenzhen Goodix Technology Co Ltd (603160.SS)	U (07/14/2025)	Rmb47.29
Winbond Electronics Corp (2344.TW)	O (05/28/2026)	NT\$177.00
WPG Holdings (3702.TW)	O (03/16/2026)	NT\$111.00
WT Microelectronics Co. Ltd. (3036.TW)	O (01/27/2026)	NT\$200.50

Duan Liu		
Dosilicon Co Ltd (688110.SS)	U (09/06/2024)	Rmb116.10
Shenzhen Longsys Electronics Co Ltd (301308.SZ)	E (02/25/2026)	Rmb364.90
Tiffany Yeh		
AllRing Tech Co. (6187.TWO)	O (09/23/2025)	NT\$1,205.00
FOCI Fiber Optic Communications Inc (3363.TWO)	O (01/15/2025)	NT\$567.00
Himax Technologies Inc (HIMX.O)	E (02/04/2026)	US\$13.44
Hon Precision (7769.TW)	O (04/17/2026)	NT\$6,255.00
MPI Corporation (6223.TWO)	O (04/17/2026)	NT\$5,565.00
Silicon Motion (SIMO.O)	O (05/06/2024)	US\$243.18
WinWay Technology Co Ltd (6515.TW)	O (04/17/2026)	NT\$6,145.00

Stock Ratings are subject to change. Please see latest research for each company.

* Historical prices are not split adjusted.

INDUSTRY COVERAGE: Japan Semiconductors

COMPANY (TICKER)	RATING (AS OF)	PRICE* (08/24/2026)
Kazuo Yoshikawa, CFA		
HOYA (7741.T)	E (10/30/2023)	¥24,895
KIOXIA Holdings (285A.T)	O (08/12/2025)	¥50,930
Renesas Electronics (6723.T)	O (11/26/2019)	¥3,428
Rohm (6963.T)	++	¥4,669
Socionext (6526.T)	U (01/27/2026)	¥1,926

Stock Ratings are subject to change. Please see latest research for each company.

* Historical prices are not split adjusted.

INDUSTRY COVERAGE: Semiconductors

COMPANY (TICKER)	RATING (AS OF)	PRICE* (08/24/2026)
Joseph Moore		
Advanced Micro Devices (AMD.O)	E (06/09/2024)	US\$456.75
Aeva Technologies Inc (AEVA.O)	E (07/19/2021)	US\$16.46
Allegro Microsystems Inc (ALGM.O)	O (02/13/2026)	US\$35.63
Ambarella Inc (AMBA.O)	O (03/29/2016)	US\$70.00
Amkor Technology Inc (AMKR.O)	E (11/08/2023)	US\$47.90
Analog Devices Inc. (ADI.O)	O (11/16/2023)	US\$371.17
Astera Labs Inc (ALAB.O)	O (05/11/2025)	US\$277.53
Broadcom Inc. (AVGO.O)	O (06/09/2024)	US\$358.76
Cerebras Systems (CBRS.O)	O (06/08/2026)	US\$185.43
GlobalFoundries Inc (GFS.O)	E (10/28/2024)	US\$45.95
Intel Corporation (INTC.O)	E (02/22/2023)	US\$87.26
IonQ Inc (IONQ.N)	E (04/25/2023)	US\$41.06
Marvell Technology Group Ltd (MRVL.O)	E (09/14/2015)	US\$229.29
Microchip Technology Inc. (MCHP.O)	E (07/10/2024)	US\$74.21
Micron Technology Inc. (MU.O)	O (10/06/2025)	US\$910.43
Navitas Semiconductor Corp (NVT.S.O)	U (04/06/2025)	US\$12.23
NVIDIA Corp. (NVDA.O)	O (03/16/2023)	US\$208.48
NXP Semiconductor NV (NXPI.O)	O (02/11/2025)	US\$222.44
ON Semiconductor Corp. (ON.O)	++	US\$71.93
Qorvo Inc (QRVO.O)	E (10/28/2025)	US\$95.37
Qualcomm Inc. (QCOM.O)	E (06/24/2026)	US\$158.53
Quantinuum (QNT.O)	E (06/29/2026)	US\$52.07
SanDisk Corporation. (SNDK.O)	O (03/03/2025)	US\$1,493.12
Semtech Corp. (SMT.C.O)	E (04/06/2025)	US\$120.91
Silicon Laboratories Inc. (SLAB.O)	E (01/19/2021)	US\$218.08

Skyworks Solutions Inc (SWKS.O)	E (11/28/2018)	US\$66.86
Texas Instruments (TXN.O)	U (04/13/2020)	US\$258.94
Wolfspeed, INC (WOLF.N)	NR (04/06/2025)	US\$24.76
Lee Simpson		
Arm Holdings plc (ARM.O)	E (04/07/2026)	US\$238.78
Cadence Design Systems Inc (CDNS.O)	O (02/14/2024)	US\$315.92
Synopsys Inc. (SNPS.O)	E (02/27/2026)	US\$394.51

Stock Ratings are subject to change. Please see latest research for each company.
* Historical prices are not split adjusted.